A pre-deployment three-way triage gate for visual-inspection anomaly detectors: escaped-defect and false-reject guarantees on MVTec AD, VisA, and MPDD

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Abstract

Anomaly detectors can rank defects well without defining a safe operating decision. We present *inspect-gate*, a pre-deployment triage layer evaluated on two detector architectures and three industrial-inspection benchmarks. It converts anomaly scores into per-category auto-pass, auto-reject, and human-defer regions with finite-sample split-conformal bounds on escaped-defect and false-reject rates. Categories whose calibration pools cannot support a requested bound are reported as *audited-not-certified* and routed to review. At the stated operating point ($\alpha_{\mathrm{miss}} = 0.10$, $\alpha_{\mathrm{fr}} = 0.05$), the gate reduces false rejects from 16.2% to 3.0% on VisA at equal escaped-defect protection, and from 36.9% to 0.0% on MPDD by deferring 73.1% of images; these are high-review diagnostic operating points, not universal economic recommendations. Escaped-defect certification holds for all 15 MVTec AD and 12 VisA categories, whereas all 33 benchmark categories refuse at $\alpha_{\mathrm{miss}} = 0.01$ because none has the required 99 defective calibration examples. A synthetic corruption stress test finds that good-score drift screening misses 34% of accepted PatchCore cells above the escaped-defect target; a defect-score marginal test catches only about 5% of those residual cells. The evidence supports calibration-data requirements and explicit refusal, not production-line deployment: all three benchmarks are academic and temporal drift remains untested.

Keywords: industrial inspection, conformal prediction, anomaly detection, selective prediction, MVTec AD, VisA

1. Introduction

Automated visual inspection is one of the most mature industrial applications of computer vision, and unsupervised anomaly detection (AD) on the MVTec AD benchmark [1] is its standard proving ground. Modern

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detectors — memory-bank methods such as PatchCore [2] and reconstruction/transformer methods such as Dinomaly [3] — report image-level area under the receiver-operating-characteristic curve (AUROC) above 0.98 and, for the newest models, above 0.99. On the usual reading the problem looks solved.

It is not solved as a decision. Image-AUROC is a threshold-free ranking summary: it says a detector orders defective above good images well, but it names no operating point, and on a real line an operating point is exactly what is needed. Worse, the two mistakes a threshold can make are not symmetric. Passing a defective part (an *escaped defect*) can ship a fault to a customer; rejecting a good part (a *false reject*) wastes yield and, at high rates, risks eroding operator trust in the system. A useful inspection system must bound both, per product category, with a guarantee that holds in finite samples rather than on average over an idealized distribution.

We take the position that the pre-deployment object is not a better detector but a certified *triage layer* on top of an existing one. Concretely, we route each image into three actions: *auto-pass* (confident-good, ship without review), *auto-reject* (confident-defective, scrap or rework without review), and *defer* (send to a human). The value proposition is that the two automated actions carry certificates: the auto-pass region is calibrated so that the probability a truly defective image lands there is at most α_{miss}, and the auto-reject region so that the probability a truly good image lands there is at most α_{fr}; everything else defers. The human-review budget is then whatever the data forces, and — crucially — it is visible: a category whose data cannot support the requested guarantee defers more, rather than pretending to a certificate it cannot honor. We evaluate the layer on three complementary benchmarks by design: MVTec AD, whose small per-category good pools stress the refusal machinery; VisA, whose lower score ceilings and larger good pools let both certificates bind everywhere and give the deferral-skill audit real headroom; and MPDD (real painted-metal-parts fabrication), whose uniformly small good pools make it the stingiest certifiability point — false-reject certifiable in 0/6 categories under the primary protocol, the extreme that shows the gate refuses precisely when the data cannot pay for a certificate.

This framing turns a benchmark score into an auditable decision and makes three contributions.

- **A certified decision-knowledge layer for inspection.** The deliverable is a per-category *certified operating envelope*: an explicit, machine-interpretable representation of the plant's acceptance policy — calibrated thresholds, certified error budgets on two axes (escaped-defect and false-reject), and refusal states — consumed downstream by the quality-control workflow. The contribution is not a new estimator — both guarantees reduce exactly to split-conformal quantiles (Section 3.1: false-reject directly, escaped-defect through a score-negation order-statistic identity) — but the coupled system assembled from them: explicit per-category certifiability floors, first-class refusal, and an auditable decision layer over any backbone's scores. To our knowledge no published method delivers a coupled two-axis certified operating envelope with first-class refusal on MVTec AD or VisA; Section 5.7 makes the coupled system's added value concrete rather than asserted: recomputed on the identical score dumps, the single-threshold construction our own escaped-defect certificate specializes (conformal risk control) pays 36.9% false-reject on MPDD to hold the same escaped-defect rate our dual gate holds at 0.0%, purely by auto-rejecting the ambiguous band instead of deferring it. We foreground the "no new math" property deliberately: it is what makes the guarantees finite-sample-exact and the tool a thin, testable wrapper rather than a heuristic.

- **Refusal as a first-class outcome, with teeth.** A per-category certifiability floor $\alpha_{\min} = 1/(n_{\text{cal}} + 1)$ makes "we cannot certify this category at this rate" an explicit, reported state (audited-not-certified), never a silently loosened threshold. This is not cosmetic: on our own substrate (Table A.1), a non-refusing baseline that issues an auto-reject threshold and claims the requested false-reject rate $\alpha_{\text{fr}} = 0.05$ in the 11 categories the gate refuses would be asserting a guarantee the calibration pool cannot support — the achievable floor there is 0.06–0.14 (e.g. toothbrush $1/7 = 0.14$), 1.2×–2.9× looser than claimed — so it silently over-promises where the gate refuses and defers. (On the escaped-defect axis the floor clears $\alpha_{\text{miss}} = 0.10$ in all 15 categories, so this failure mode does not arise there; a post-hoc binding demonstration (Section 5.12) confirms this empirically — in a range of (backbone, category) cells a non-refusing fixed-threshold baseline's *realized* escaped-defect or false-reject rate exceeds the target while the certified gate holds within it on the same data.)

- **A practitioner-facing audit.** Beyond the certificate, we test whether field-standard threshold practice earns any deferral skill over honest random deferral (an excess area-under-the-risk–coverage-curve (AURC) audit), reporting a constructive or a debunking verdict symmetrically; the preregistered confirmatory pooled audit returns the constructive verdict — practice earns real deferral skill in all five seeds (Section 5.15).

Figure 1 summarizes the resulting pipeline, from a frozen backbone's per-image score through the G1/G2 certificates and certifiability floor to the three-way auto-pass/defer/ auto-reject route.

We state the evidence boundary explicitly. The preregistration was frozen before confirmatory analysis; MVTec AD is the confirmatory benchmark,

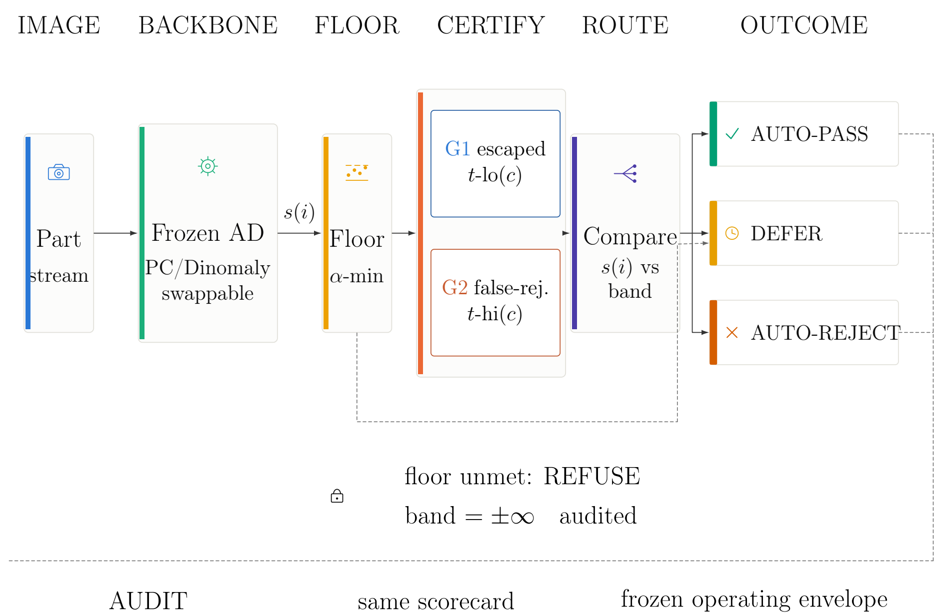

Figure 1: Horizontal method overview. A frozen, interchangeable PatchCore or Dinomaly backbone scores each image. Per-category calibration first checks the certifiability floor $\alpha_{\min} = 1/(n_{\text{cal}} + 1)$ (Section 3.2); when met, the parallel G1 and G2 cards set $t_{\text{lo}}(c)$ and $t_{\text{hi}}(c)$ so that $P(s \le t_{\text{lo}} \mid \text{defective}) \le \alpha_{\text{miss}}$ and $P(s \ge t_{\text{hi}} \mid \text{good}) \le \alpha_{\text{fr}}$, respectively (Section 3.1). Comparing s_i with this band routes the part to auto-pass, defer, or auto-reject. The dashed floor-control branch is distinct from that data path: an unmet floor refuses the axis (band $= \pm\infty$), records audited-not-certified, and forces defer rather than auto-reject. All three outcomes feed the same audit scorecard and frozen operating envelope; the schematic makes no claim that either backbone is superior.

while VisA and MPDD are exploratory additions run under the same protocol. The frozen G2 train-holdout remedy is reported separately, and no uncomputed result is used in the claims.

1.1. A portable operating envelope for the inspection line

The gate's output is a deployable operating envelope: for every product category it materialises explicit, machine-readable operating parameters — the two calibrated thresholds $(\lambda_{\text{pass}}, \lambda_{\text{reject}})$, the certified error budgets $(\alpha_{\text{miss}}, \alpha_{\text{fr}})$ they enforce, and, where calibration data cannot support the request, an *audited-not-certified* refusal state — which downstream quality-control logic (routing, re-inspection staffing, audit trails) consumes directly. The envelope is backbone-agnostic and portable across lines: swapping the detector changes the scores, not the schema of the envelope, and the refusal state is an explicit operating signal rather than an implementation detail. The academic benchmarks used here validate this calibration-and-refusal machinery; deploying the same envelope schema on production-line data with temporal drift is the explicit next step (Section 7).

2. Related work

Distribution-free risk control.. The gate is an application of conformal prediction [4, 5, 6]: given an exchangeable calibration set, an order statistic of the calibration scores yields a threshold with a finite-sample coverage guarantee. The specific ingredient we use is the split (inductive) conformal construction [7], whose single calibration/estimation split makes it cheap enough to apply per category at deployment time. Set-valued and adaptive variants of the same idea — least-ambiguous set-valued classifiers with a bounded error level [8] and adaptive-coverage prediction sets [9] — show how a one-sided error target maps onto a calibrated threshold, which is exactly the primitive our two certificates specialise. The broader risk-controlling line — risk-controlling prediction sets [10] and Learn-then-Test [11] — generalizes this to bounding arbitrary monotone risks and to multiple-hypothesis calibration. All of these guarantees are conditional on exchangeability between calibration and deployment data; conformal prediction under covariate shift [12] characterises what breaks when that assumption fails, and is the lens through which we frame threshold-transfer across production periods as future work (Section 7). Our escaped-defect and false-reject bounds are the two one-sided special cases that split conformal already delivers exactly (Section 3.1); we deliberately use the simplest sufficient construction rather than the more general risk-controlling prediction sets / learn-then-test (RCPS/LTT) machinery, and cite the latter as the family the method belongs to.

Industrial anomaly detection.. MVTec AD [1] defined the per-category unsupervised setting we use, and the field it seeded is now broad enough to warrant dedicated surveys [13]. The dominant families all emit an image- or pixel-level anomaly score our gate can consume unchanged: memory-bank and feature-distribution methods such as PatchCore [2], PaDiM [14], and SPADE [15]; student–teacher and knowledge-distillation methods such as uninformed students [16] and reverse distillation [17]; reconstruction and synthetic-defect methods such as DRÆM [18] and CutPaste [19]; normalizing-flow methods such as FastFlow [20]; and recent simplified or transformer models such as SimpleNet [21] and Dinomaly [3]. Of these, PatchCore and Dinomaly are the two reproduced backbones here; EfficientAD [22] is a fast alternative we discuss but do not certify (its only public implementations are community reimplementations, so it cannot anchor a binding reproduction gate). Dinomaly descends from the multi-class (one model, many categories) line initiated by UniAD [23] — the setting its published VisA figure reports and the one we use on VisA (Section 4). In the applied inspection literature, this detector line has been carried directly into deployable inspection systems: component-aware anomaly detection for logical industrial inspection [24], defect-aware transformer networks for visual surface-defect detection [25], and autoencoder-based surface anomaly inspection [26], alongside segmentation-based deep-defect detection [27] and broader automated-visual-inspection surveys [28]. That body of work optimises the *detector*; our contribution is the orthogonal decision layer that sits on top of any such detector's scores and attaches a certified routing guarantee. MPDD [29] adds a real painted-metal-parts fabrication benchmark whose non-homogeneous backgrounds and small good pools make it a deliberately harsher industrial-realism stress test than the two academic staples — though, like them, still an academic benchmark rather than production-line data (see Limitations, Section 7). We use anomalib [30] for the PatchCore implementation. Our contribution is orthogonal to all of these: we consume their scores and add a decision layer, and we improve none of them.

Selective prediction and abstention.. The defer action is a selective-prediction mechanism [31, 32], but applied at the decision layer with a two-sided, per-category certificate rather than a single accuracy-coverage curve or a learned reject head; the auto-reject certificate (false-reject control) has no analogue in standard selective classification, which abstains only around the positive decision.

Conformal anomaly detection and abstention.. Conformal methods have been applied to anomaly detection before, but not to the certified inspection-triage decision we target. Early work bounded false-alarm rates for sequential and trajectory anomalies [33, 34]; recent tooling controls the false-discovery rate over flagged anomalies [35]; conformal risk control [36] and its graph instantiation CRC-SGAD [37] bound a monotone risk (e.g. false-negative rate/false-positive rate, FNR/FPR) but emit a single flag, not a three-way route; and dual-threshold conformal abstention for autonomous-system perception [38] adds a reject option on camera/LiDAR (and classification) tasks, with no industrial-inspection estimand and no evaluation on MVTec AD. Stated plainly, our delta over that dual-threshold line is not conformal machinery but the inspection instantiation: the two error axes named and certified as escaped-defect and false-reject budgets, per-category certifiability floors with first-class refusal, and the audited deployment protocol around them. The closest defect-detection application [39] controls a mean error rate for pixel-level surface-defect segmentation, using recall/false-discovery-rate (FDR)-style losses, whereas this paper operates on image-level anomaly scores and certifies a three-way triage with abstention plus explicit per-category sample-size certifiability floors. It therefore differs in substrate, route, and the operating-state refusal rule; it does not claim pixel-level segmentation control. no published method delivers a certified *three-way* triage (auto-pass / auto-reject / defer) on MVTec AD with *coupled* escaped-defect and false-reject guarantees and explicit per-category certifiability floors; the preregistered K6 scoop gate, run as a fresh pre-submission re-scan after the freeze, confirms this — the closest published systems [38, 37, 39] each differ on a load-bearing axis (substrate, estimand coupling, or the defer option), and two adjacent 2026 works (a selective-risk certificate on generic vision, and a conformal cyber-physical-systems detector on sensor time-series) target neither the image-inspection estimand nor the three-way route. That is the seam this paper occupies. Section 5.7 instantiates the single-threshold CRC construction [36] as a quantitative baseline on our own canonical score dumps under the identical protocol, isolating what the three-way route and its coupled false-reject certificate add over the nearest single-flag alternative. Our validation is on academic benchmarks; deployment on production lines with temporal drift and non-stationary defect arrival is future work (Section 7).

3. Methods

Table 1 collects the short-codes reused throughout Sections 3–7, as a single reference point instead of a re-derivation on each reuse.

Table 1: Notation and short-code glossary. Every code below recurs at least twice in Sections 3–7; consult this table rather than re-deriving a code's meaning from context on each reuse.

Code	Meaning
G1	Certified escaped-defect bound (Section 3.1)
G2	Certified false-reject bound (Section 3.1)
K1–K7	Kill-gates: coverage sanity, vacuity, backbone floor ($\geq$ 0.90 AUROC), audit headroom, calibration floor, prior-work re-scan, compute guard (in order)
V1 tier-1	Validity audit: certificate-validity statistic
V1 tier-2	Validity audit: stringent Clopper–Pearson estimator-variance readout
A1–A3	Amendments: (A1) tier-2 power floor graded per repeat, not pooled; (A2) near-target tier-2 estimator variance is expected, not failure; (A3) false-reject axis graded only over G2-certified cells
B1/B2/B3	Threshold practices: (B1) single global fixed threshold; (B2) per-category tuned threshold; (B3) train-good quantile threshold
C2	Confirmatory excess-AURC audit (pooled Holm family, Section 5.15)
CRC	Conformal Risk Control, the single-threshold conformal baseline

3.1. The three-way gate and its two certificates

Each image i in category c has an anomaly score s_i with the convention *higher = more anomalous*. The gate is a pair of per-category thresholds $(t_{\text{lo}}(c), t_{\text{hi}}(c))$ with $t_{\text{lo}} \leq t_{\text{hi}}$: images with $s_i \leq t_{\text{lo}}(c)$ auto-pass, images with $s_i \geq t_{\text{hi}}(c)$ auto-reject, and the rest defer. (If a refusal or crossed thresholds leave the auto-reject band empty, the overlap simply defers, and both guarantees still hold.)

Equivalently, Equation (1) reports the decision for image i:

$$D(s_i; c) = \begin{cases} \text{pass}, & s_i \leq t_{\text{lo}}(c), \\ \text{reject}, & s_i \geq t_{\text{hi}}(c), \\ \text{defer}, & t_{\text{lo}}(c) < s_i < t_{\text{hi}}(c). \end{cases} \tag{1}$$

G2 — certified false-reject, directly.. $t_{\mathrm{hi}}(c)$ is the split-conformal upper quantile of the *good* calibration scores at level α_{fr}: with $n_{\mathrm{cal}}^{\mathrm{good}}$ good calibration scores, Equation (2) sets it to the order statistic at rank $\lceil (n+1)(1-\alpha_{\mathrm{fr}})\rceil$, guaranteeing $P(s \geq t_{\mathrm{hi}} \mid \mathrm{good}) \leq \alpha_{\mathrm{fr}}$ under exchangeability as in Equation (3). This is exactly the split-conformal upper-quantile construction applied to the good scores.

$$t_{\mathrm{hi}}(c) = S^{\mathrm{good}}_{(k_{\mathrm{hi}})}, \qquad k_{\mathrm{hi}} = \left\lceil \left(n_{\mathrm{cal}}^{\mathrm{good}} + 1\right)\left(1 - \alpha_{\mathrm{fr}}\right) \right\rceil, \tag{2}$$

$$\Pr\{s \geq t_{\mathrm{hi}}(c) \mid \mathrm{good},\, c\} \leq \alpha_{\mathrm{fr}}. \tag{3}$$

G1 — certified escaped-defect, by negation.. Bounding $P(s \leq t_{\mathrm{lo}} \mid \mathrm{defective})$ is the mirror image: apply the same split-conformal construction to the *negated* defective calibration scores $Y = -X$ at level α_{miss}, then map the resulting upper quantile back through the negation. The gate module proves the order-statistic identity in Equation (4), and it is checked against the design's $\lfloor \alpha(n+1) \rfloor$ formula in the accompanying unit tests. No new estimator is introduced; both certificates are the same conformal quantile.

$$\begin{gathered} t_{\mathrm{lo}}(c) = -Y_{(k_{\mathrm{lo}})}, \qquad Y = -X, \qquad k_{\mathrm{lo}} = \left\lceil \left(n_{\mathrm{cal}}^{\mathrm{def}} + 1\right)\left(1 - \alpha_{\mathrm{miss}}\right) \right\rceil, \\ \Pr\{s \leq t_{\mathrm{lo}}(c) \mid \mathrm{defective},\, c\} \leq \alpha_{\mathrm{miss}}. \end{gathered} \tag{4}$$

Mondrian per-category stratification.. Calibration is stratified by category (a Mondrian conformal construction): each category gets its own $(t_{\mathrm{lo}}, t_{\mathrm{hi}})$, so the guarantee is conditional on category, not marginal. An optional finer per-defect-type stratification is exploratory only (Section 7).

Where the labeled defectives come from.. G1 calibrates on labeled *defective* images per category ($n_{\mathrm{cal}}^{\mathrm{def}}$ in Table A.1) — a requirement unsupervised anomaly detection is deployed specifically to avoid needing at detector-training time, since labeled defects are scarce, especially early in a line's life. This is a real cost relative to the unsupervised premise: the certificate is only as good as the labeled defective pool behind it, and unlike the detector itself that pool cannot be assembled from good parts alone. In deployment it is not a fixed benchmark quantity but accrues over time, from rework, scrap, and quality-escape logs as the line runs. The $\alpha_{\min} = 1/(n_{\mathrm{cal}}^{\mathrm{def}} + 1)$ floor and the refusal rule below are what make this honest rather than aspirational: a category with a thin defect history is refused and reported audited-not-certified rather than issued a certificate its data cannot support, and MPDD's connector category already demonstrates the mechanism in practice ($n_{\mathrm{cal}}^{\mathrm{def}} = 7 < 9$ at $\alpha_{\mathrm{miss}} = 0.10$, refused at the frozen rate; Section 5.3).

3.2. Certifiability floors and refusal

Split conformal cannot certify an arbitrarily small rate from a finite calibration pool: the smallest achievable level is $\alpha_{\min} = 1/(n_{\mathrm{cal}} + 1)$. A category is certifiable for a guarantee iff $\alpha_{\min} \leq \alpha$ as in Equation (5); otherwise the gate *refuses*, setting the corresponding threshold to $\pm\infty$ (the band is empty, all such images defer) and marking the category audited-not-certified for that axis. Refusal is emitted, never rounded away:

$$\alpha_{\min}(n_{\text{cal}}) = \frac{1}{n_{\text{cal}} + 1}, \qquad \text{certifiable at level } \alpha \iff \alpha_{\min}(n_{\text{cal}}) \leq \alpha. \quad (5)$$

This is the honesty rule that makes the floor table (Section 5.3) a result rather than a limitation.

The gate's mechanism is made concrete on one repeat of the frozen protocol in Figure 2.

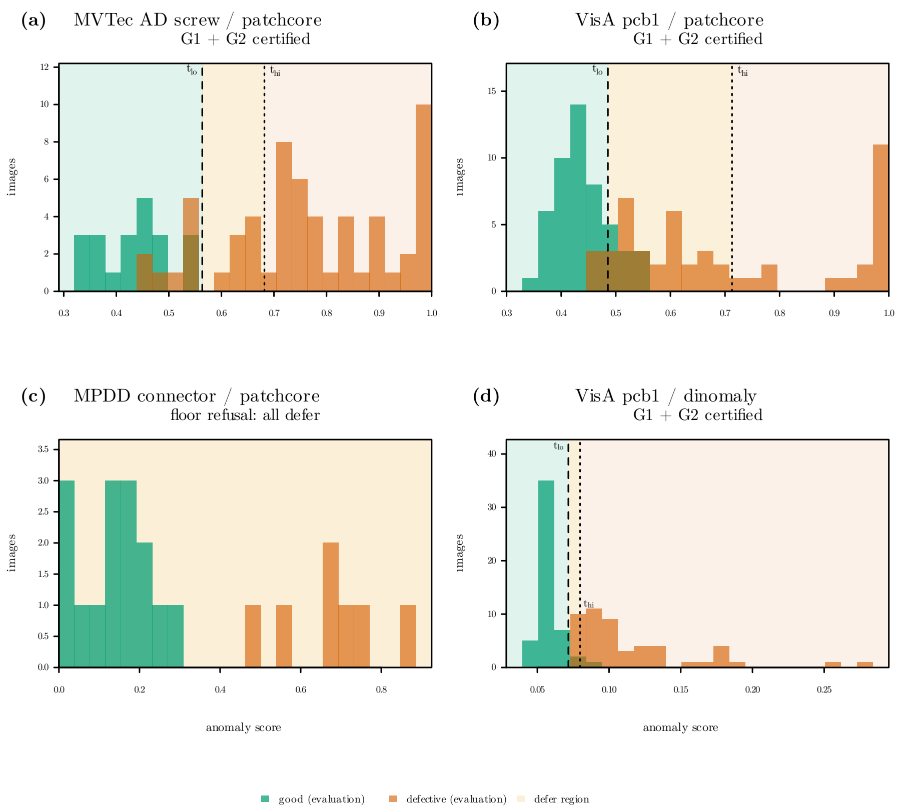

Figure 2: The gate's mechanism on real scores (seed 0, repeat 0; evaluation-half histograms, thresholds calibrated on the calibration half). **a:** MVTec AD screw (PatchCore): both certificates, defer band absorbs the overlap region. **b:** VisA pcb1 (PatchCore): both certified with a wider band. **c:** MPDD connector (PatchCore): separation looks clean to the eye yet the gate refuses — pools too small for the $1/(n_{\text{cal}} + 1)$ floors, both thresholds at $\pm\infty$. **d:** VisA pcb1 under *Dinomaly* (contrast with b): same category, different scorer; both G1 and G2 certify with a narrow band ($t_{\text{lo}} = 0.072$, $t_{\text{hi}} = 0.080$), showing that a different detector with cleaner score separation can close the defer band substantially.

Figure 2 makes the mechanism concrete on one repeat of the frozen protocol: where the two score masses are well separated the band is narrow and both certificates issue (panel a); where they overlap the band widens to cover the overlap and deferral rises (panel b); and where calibration pools are too thin the gate refuses outright (panel c) — the refusal reads the pool sizes, not the apparent separability, which is precisely why a threshold picked by eye on panel c would be an uncertified decision the gate declines to automate.

3.3. Validity audit (V1) and kill-gates (K1–K7)

We fix $\alpha_{\text{miss}} = 0.10$ and $\alpha_{\text{fr}} = 0.05$ throughout, with a +3pp estimator tolerance (thresholds 0.13 / 0.08) and 95% one-sided Clopper–Pearson intervals [40]. The *validity audit* V1 has two tiers. Tier-1 checks that the realized mean escaped-defect and false-reject rates in Equation (6) lie within target+3pp over repeated calibration/evaluation splits; this is the certificate-validity statistic. Tier-2 forms the Clopper–Pearson upper bound in Equation (7) and is graded, per the frozen amendments, as a stringent estimator-variance readout (Section 5.16), because on MVTec AD its power floor is not met per repeat and its tolerance is near-unpassable by construction for a correctly-calibrated gate. The manuscript-level V1 family contains $33 \times 2 \times 5 = 330$ benchmark–category–backbone–seed cells (MVTec AD 15, VisA 12, MPDD 6 categories; two backbones; five seeds). We therefore keep the confirmatory inferential claims separate from this diagnostic grid: V1 reports deterministic pass/fail counts, finite-sample power exclusions, and one-sided Clopper–Pearson intervals, while the only hypothesis-test family used to claim practice skill is the C2 Holm-adjusted excess-AURC audit (Section 5.15). Post-freeze VisA/MPDD V1 cells are labelled exploratory and are not pooled into the frozen MVTec confirmatory family.

$$\begin{aligned} \widehat{r}_{\text{miss}} &= \frac{1}{|\mathcal{E}_{\text{def}}|} \sum_{i \in \mathcal{E}_{\text{def}}} \mathbf{1}\{D(s_i; c_i) = \text{pass}\}, \\ \widehat{r}_{\text{fr}} &= \frac{1}{|\mathcal{E}_{\text{good}}|} \sum_{i \in \mathcal{E}_{\text{good}}} \mathbf{1}\{D(s_i; c_i) = \text{reject}\}. \end{aligned} \tag{6}$$

$$U_{\text{CP}}(k, m; \gamma) = \begin{cases} \text{Beta}^{-1}_{1-\gamma}(k+1, m-k), & k < m, \\ 1, & k = m, \end{cases} \qquad \gamma = 0.05. \tag{7}$$

A battery of preregistered kill-gates guards the construction: K1 (coverage sanity — halt if tier-1 fails in ≥ 5 of the $15 \times B$ cells), K2 (vacuity — kill if median deferral > 0.80 in $\geq 8/15$ categories on all backbones), K3 (backbone floor — mean image-AUROC ≥ 0.90), K4 (audit headroom), K5 (calibration floor — fires only if $\alpha_{\min} > 0.10$ in ≥ 3 categories), K6 (a citation re-scan for a prior certified triage), and K7 (a compute guard).

3.4. Excess-AURC audit

Beyond validity, we ask whether standard thresholding practice earns deferral skill at all. For each practice — B1 (global fixed threshold), B2 (per-category tuned), B3 (train-good quantile) — we compare its area-under-the-risk-coverage-curve against the closed-form random-deferral null (evaluated in closed form, never Monte-Carlo'd), with the statistic in Equation (8), using a matched-abstention, category-stratified permutation p-value (2000 permutations), a category-blocked bootstrap CI, and a roster-derived Holm [41] correction. The verdict is symmetric: if no practice beats random the paper reports a debunking finding; if all do it reports a constructive one.

$$\begin{aligned}
\widehat{C}_\pi(\tau) &= \frac{1}{|\mathcal{E}|}\sum_{i\in\mathcal{E}} A_{\pi,\tau}(i),\\
\widehat{R}_\pi(\tau) &= \frac{\sum_{i\in\mathcal{E}} L_i A_{\pi,\tau}(i)}{\sum_{i\in\mathcal{E}} A_{\pi,\tau}(i)},\\
\widehat{\mathrm{AURC}}(\pi) &= \int_0^1 \widehat{R}_\pi(u)\,du,\\
\widehat{\Delta}_{\mathrm{AURC}}(\pi) &= \widehat{\mathrm{AURC}}(\pi) - \widehat{\mathrm{AURC}}(\text{random deferral}).
\end{aligned} \tag{8}$$

4. Experimental setup

Data.. Three public benchmarks. MVTec AD [1]: 15 object/texture categories, 3,629 train-good images, and a test set of 467 good + 1,258 defective = 1,725 images across 73 defect types. VisA [42], under the official one-class `spot-diff` split: 12 categories, 8,659 train-good images, and a test set of 962 good + 1,200 anomalous = 2,162 images (VisA's public ground truth carries a single "bad" stratum per category, so the defect-type Mondrian axis is unavailable there). MPDD [29] (post-freeze exploratory; see below): 6 real painted-metal-part categories, 888 train-good images, and a test set of 176 good + 282 defective = 458 images from a public release mirrored for staging. Splits follow each benchmark's official protocol, or a frozen per-category test manifest where none ships; score records are joined to those splits with refusal on any missing or extra image.

Figure 3 illustrates the three-way decisions on real parts across the route and ground-truth classes.

Backbones (Branch A, $B = 2$).. PatchCore [2] via anomalib [30] (a Wide ResNet-50-2 backbone, features from the second and third stages, coreset ratio 0.10)[1] and Dinomaly [3] (canonical score dump, its own eval script; on VisA, the reference repo's multi-class setting — one model per seed for all 12 categories — which is the setting its published VisA figure reports). Both were confirmed present and reproduction-passing before any calibration, so the preregistered two-backbone branch (Branch A) is active; the same two backbones and configurations are used unchanged on VisA.

[1]Target 0.991 is PatchCore's published 25%-coreset mean; we score the 10%-coreset point (PREREG D3). See Table 2.

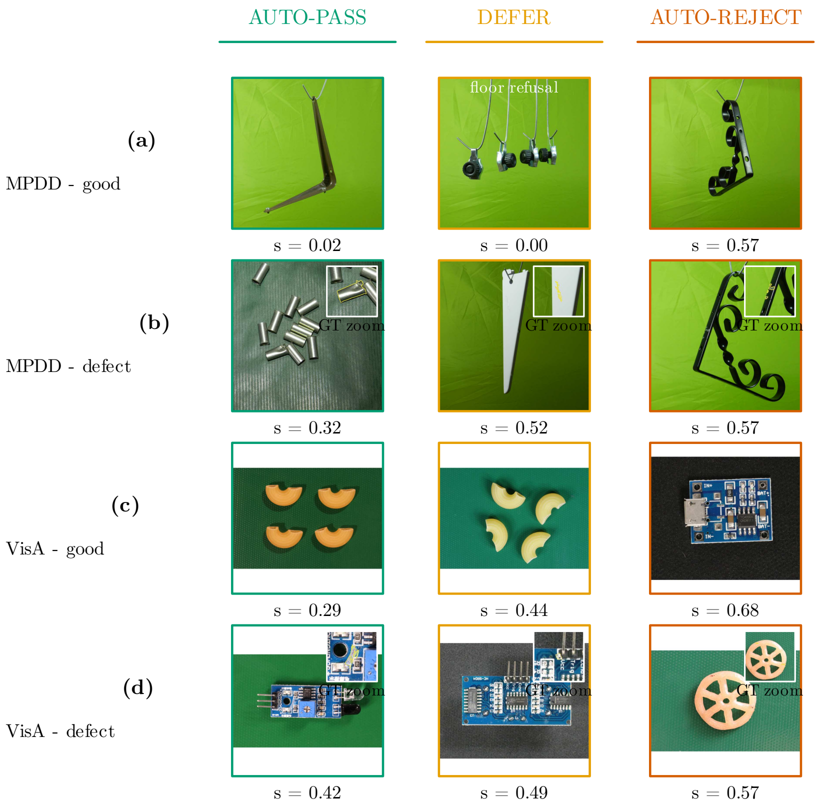

Figure 3: Three-way decisions on real parts (PatchCore, seed 0, repeat-0 split; illustrative, not a new experiment). Columns show auto-pass, defer, and auto-reject; rows show MPDD good, MPDD defective, VisA good, and VisA defective examples. Border color encodes the route and each tile gives its anomaly score. Insets and yellow contours use dataset ground truth to locate defects and are not model localization. MPDD uses the train-holdout rescue arm (Section 5.11); VisA uses the primary protocol. The connector defer tile is marked *floor refusal*: connector's escaped-defect side cannot certify at $\alpha_{\text{miss}} = 0.10$ ($\alpha_{\text{min}} = 0.125$), so $t_{\text{lo}} = -\infty$ and every image defers — a different decision class from a within-band defer, which the tile now names explicitly. Together the examples expose all six ground-truth/route classes, including false rejection and escaped-defect cases.

Protocol.. Primary protocol: test-side good calibration, no train-holdout. For each (backbone, category, seed) we draw $R = 20$ stratified 50/50 calibration/evaluation splits (split-seed = repeat index), with 5 backbone seeds (0–4).

5. Results

All numbers in this section are recomputed from the frozen local score dumps by the archived analysis pipeline — a MVTec AD analysis stage, a stage-identical VisA analysis stage, a stage-identical MPDD analysis stage (with its own train-holdout rescue stage), and the G2-promotion stage, each emitting a frozen result summary; none is trusted from a training log or the literature.

5.1. Reproduction gate

On all three benchmarks the gate recomputes image-AUROC (Mann–Whitney rank-sum identity) directly from the local scores and compares it against a named published target (Table 2). On MVTec AD both backbones pass in all 5 seeds, and K3 (mean AUROC ≥ 0.90) holds with wide margin. On VisA, Dinomaly passes in all 5 seeds against its own published VisA figure (0.987, multi-class setting), and our recomputation again reproduces its reported per-seed mean image-AUROC seed for seed (max per-category deviation 3.3×10^{-5} over all 60 seed×category cells). PatchCore has no repo-confirmed published VisA figure, so per the tool's no-guessed-target rule its VisA row is *descriptive* (no pass/fail is graded); its recomputed mean of 0.903–0.908 still clears the K3 floor, though with thin margin (seed 3: 0.9033, i.e. 0.33pp above the 0.90 floor), and the drop from 0.982 (MVTec) is exactly the lower-ceiling substrate the audit needs (Section 5.14). On MPDD, Dinomaly passes in all 5 seeds against its published MPDD multi-class figure (0.972; recomputed mean 0.9712–0.9738, grand mean 0.9722 ± 0.0010), our recomputation matching the box's per-category training-log AUROC to $\leq 4.9 \times 10^{-5}$; PatchCore-on-MPDD has no repo-confirmed published figure, so per the same no-guessed-target rule its row is descriptive (recomputed mean 0.948–0.957).

Table 2: Reproduction gate, 5 seeds per backbone per benchmark. Mean image-AUROC range is over seeds; the weakest category is named. "n/a": no repo-confirmed published PatchCore-on-VisA/MPDD figure exists, so that row is descriptive by design (no pass/fail graded). Recomputed from the frozen MVTec AD, VisA, and MPDD analysis summaries.

Benchmark	Backbone	Target (±0.02)	Mean I-AUROC (seed range)	Weakest category	Gate
MVTec AD	PatchCore	0.991	0.9817–0.9826	toothbrush 0.911–0.917	PASS 5/5
MVTec AD	Dinomaly	0.996	0.9957–0.9965	capsule 0.976–0.982	PASS 5/5
VisA	PatchCore	n/a	0.9033–0.9083	macaroni2 0.656–0.665	descriptive
VisA	Dinomaly	0.987	0.9866–0.9876	macaroni2 0.956–0.966	PASS 5/5
MPDD	PatchCore	n/a	0.9481–0.9568	bracket black 0.872–0.888	descriptive
MPDD	Dinomaly	0.972	0.9712–0.9738	bracket black 0.929–0.940	PASS 5/5

5.2. Cross-seed stability

PatchCore's five seeds are genuinely distinct: all 60 seed-0-vs-seed-N score-table comparisons (15 categories × 4 seeds) are 0/60 bit-identical, with max $|\Delta\text{score}|$ ranging 0.026–0.153 (mean 0.074) — a real but modest variance source, so PatchCore is reported as a 5-effective-seed backbone, not collapsed to one. Dinomaly is low-variance: mean image-AUROC across seeds 0.9960 ± 0.00029, with the widest per-category spread only 0.0060 (capsule). The same picture holds on VisA: PatchCore is 0/48 bit-identical (max $|\Delta\text{score}|$ 0.044–0.256), and Dinomaly's mean image-AUROC across seeds is 0.9870 ± 0.0004 with the widest per-category spread 0.0104 (macaroni2). On MPDD the pattern repeats once more: PatchCore is 0/24 bit-identical across the seed-0-vs-seed-N comparisons, and Dinomaly's mean image-AUROC across seeds is 0.9722 ± 0.0010.

5.3. Certifiability floors and the G1/G2 split

Table A.1 is the paper's honesty figure. The per-category calibration counts computed from the full local test arrays match the preregistered arithmetic-only floor table exactly (0/15 mismatches for both backbones, identical across all 5 seeds). At the frozen rates, escaped-defect (G1) is certifiable in all 15 categories; false-reject (G2) is certifiable in exactly 4 — cable, hazelnut, screw, transistor — the categories with $n_{\text{cal}}^{\text{good}} \geq 19$. The other 11 categories are refused for G2 and reported audited-not-certified. We are careful about what this exact match does and does not show: the per-category counts are a *deterministic* function of the fixed MVTec test split (round-half-to-even of $n/2$), so the 0/15 agreement with the preregistered table is a pipeline-correctness check — it confirms the data staging re-derives the frozen counts with no off-by-one or CSV-read error — not an empirical result about the gate. The genuinely empirical result is the V1 tier-1 validity audit below (Section 5.13), measured from real calibrate/route/evaluate runs, which could have failed had the calibration or routing pipeline been broken.

VisA: the same arithmetic, a friendlier split.. On VisA the identical floor computation lands on the opposite side of the ledger: every category's test split carries 100 anomalous and 50–101 good images, so after the 50/50 calibration split $n_{\text{cal}}^{\text{def}} = 50$ everywhere ($\alpha_{\min}^{\text{G1}} = 0.0196$) and $n_{\text{cal}}^{\text{good}} \in \{25, 30, 50\}$ ($\alpha_{\min}^{\text{G2}} =$ 0.0196–0.0385), all below $\alpha_{\text{fr}} = 0.05$. G1 and G2 both certify in 12/12 categories under the primary protocol alone — no train-holdout remedy is needed on VisA — and the floors are again identical across both backbones and all five seeds (0 mismatches). This is the refusal mechanism working in both directions: it refuses 11/15 categories on MVTec AD because the data cannot support the rate, and certifies 12/12 on VisA because it can.

MPDD: the stingy extreme of pool-size-driven refusal.. MPDD's per-category test-good pools are uniformly small (26–32 images $\rightarrow n_{\text{cal}}^{\text{good}}$ 13–16, every one below the 19 the false-reject floor needs at $\alpha_{\text{fr}} = 0.05$), so G2 is certifiable in 0/6 categories under the primary protocol, on both backbones, identical across all five seeds (0 mismatches). G1 (escaped-defect) stays healthy at 5/6 — only connector fails, its 14 test-defective yielding $n_{\text{cal}}^{\text{def}} = 7 < 9$. This places MPDD at the stingy end of three points consistent with pool-size-driven refusal, ordered by good-pool size: MPDD 0/6 $\rightarrow$ MVTec AD 4/15 $\rightarrow$ VisA 12/12 (all three computed apples-to-apples by the identical floor code path). MPDD is thus not an intermediate point but the cleanest demonstration of the paper's thesis — the certifiable count is a property of how much exchangeable good data the protocol can spend, and the gate refuses precisely when the data cannot pay for a 5%-false-reject certificate. Figure 4 makes this certify-or-refuse pattern visible at the category level across all three benchmarks: VisA certifies both axes in every category, MVTec AD's false-reject row is hatched (refused) everywhere except the four categories in Table A.1, and MPDD's false-reject row is hatched throughout — the visual complement of the same table.

5.4. The operating-point frontier (post-freeze exploratory)

The post-freeze operating-point sweep is shown in Figure 5.

The headline rates above all sit at one operating point ($\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} =$ 0.05). Because a plant will ask what a tighter escape budget costs, we sweep the full grid $\alpha_{\text{miss}} \in \{0.20, 0.10, 0.05, 0.02, 0.01\} \times \alpha_{\text{fr}} \in \{0.10, 0.05, 0.02\}$ through the byte-identical frozen calibration and routing code (Figure 5; prereg-exploratory slot F4, computed post-freeze). Three readings. First, VisA has real headroom: both axes certify in all 12 categories down to $\alpha_{\text{miss}} = 0.02$ — Dinomaly's deferral rises only 6.5% $\rightarrow$ 12.9% — before

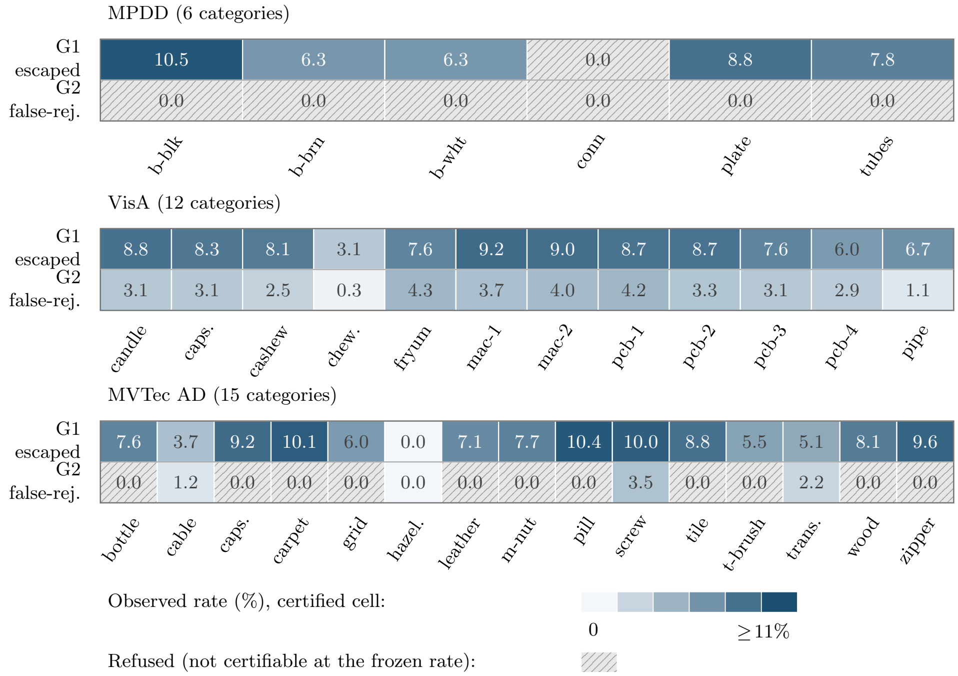

Figure 4: Per-category certification map for escaped-defect (G1) and false-reject (G2) at the frozen rates ($\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$), pooled over both backbones and five seeds, for MPDD, VisA, and MVTec AD. Colored cells are certified, shaded by the observed rate; hatched gray cells are refused as not certifiable at the frozen rate under the primary protocol (all of MPDD's G2 row, MPDD's connector on G1, and the 11/15 MVTec AD categories below the false-reject floor in Table A.1). A colored 0.0 is a certified cell with zero observed errors; a hatched 0.0 reports the observed rate but withholds the certificate because the calibration floor is unmet. MPDD codes are b-blk/bracket black, b-brn/bracket brown, b-wht/bracket white, conn/connector, and plate/metal plate. VisA abbreviations are caps./capsules, chew./chewing gum, mac-1/macaroni1, mac-2/macaroni2, pcb-1–pcb-4, and pipe/pipe fryum. MVTec AD abbreviations are caps./capsule, hazel./hazelnut, m-nut/metal nut, t-brush/toothbrush, and trans./transistor; remaining category labels are written in full.

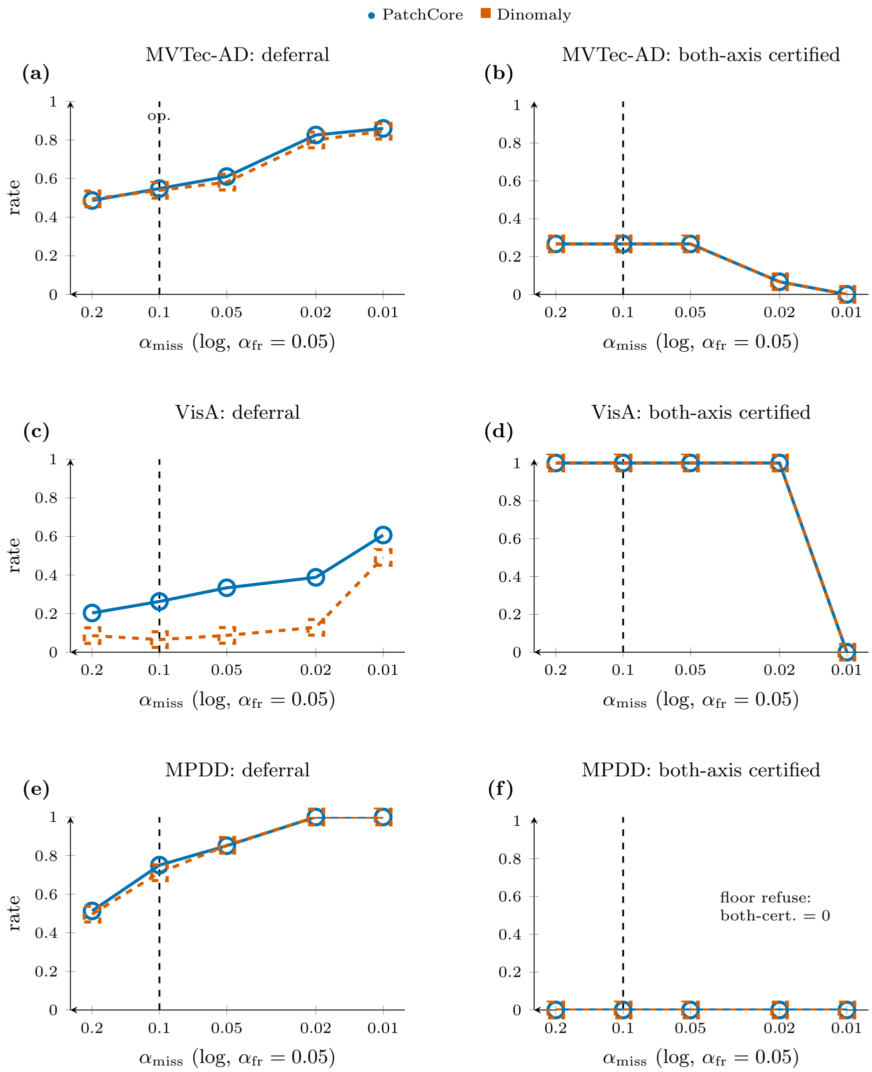

Figure 5: The operating-point frontier on the frozen code path (post-freeze exploratory sweep; the construction reproduces every frozen artifact exactly at the paper's operating point). Rows are MVTec AD (a,b), VisA (c,d), and MPDD (e,f). Left panels show mean deferral versus the escaped-defect target α_{miss} at $\alpha_{\text{fr}} = 0.05$; right panels show the fraction of categories certified on both axes. Solid circles denote PatchCore and dashed squares Dinomaly; shaded bands span seed minima to maxima. The vertical dashed line marks the paper's operating point. The flat zero in panel (f) for MPDD at $\alpha_{\text{miss}} \in \{0.02, 0.01\}$ is a measured refusal: the floor ($\alpha_{\min} = 1/(n_{\text{cal}}^{\text{frac}} + 1)$) refuses all six MPDD categories, so both-certified is identically zero on every repeat and deferral saturates at 1.0 (no silent decisions).

certification collapses to 0/12 at 0.01, exactly where the floor arithmetic ($n_{\text{cal}}^{\text{def}} \geq 99$) predicts. Second, MVTec AD's binding constraint is pool size, not tolerance: the both-axis certified fraction is pinned at 4/15 across $\alpha_{\text{miss}} \geq 0.05$ (the good-pool floors Section 5.10's remedy addresses), and deferral worsens smoothly to 0.80–0.83 at 0.02. Third, MPDD confirms the stingy extreme, certifying nothing at any grid point on the primary protocol and saturating at 100% deferral from $\alpha_{\text{miss}} = 0.02$ down. The sweep turns the single-point headline into a priced menu a deployment can choose from, and the $\alpha_{\text{miss}} = 0.01$ collapse is shown, not asserted (Section 7).

5.5. Calibration-budget sensitivity (post-freeze exploratory)

The calibration-budget sensitivity is shown in Figure 6.

A second deployment question the single-point headline leaves open is how much calibration data the gate needs. We re-run the frozen protocol at the paper's operating point while shrinking the calibration half of each split from 50% to 10% of the test set (Figure 6; post-freeze exploratory, same byte-identical code path, verified against the frozen artifacts at fraction 0.50). Three readings. First, validity is budget-invariant: the mean pooled realized escaped-defect rate stays at or below $\alpha_{\text{miss}} = 0.10$ and the mean pooled false-reject rate at or below $\alpha_{\text{fr}} = 0.05$ at *every* fraction, on every benchmark, for both backbones (worst mean: escaped 0.095, false-reject 0.039); individual repeats on small pools exceed the targets (worst single repeat: escaped 0.25 on MPDD's ∼14-defective cells), the expected finite-n fluctuation of a marginal guarantee, not a validity breach. Second, certification collapses exactly where the floor arithmetic predicts: VisA's both-axis certification holds 12/12 down to fraction 0.30–0.20 and falls to 0 at 0.15 as per-category calibration pools cross below the $1/(n_{\text{cal}} + 1)$ floors; MVTec AD's four G2-certified categories are lost already at fraction 0.30 (its good pools are the smaller pools), while its G1 side degrades gracefully 15 → 8 of 15 at fraction 0.10. Third, the degradation is into deferral, never into silent error — at fraction 0.10 MPDD defers 100% of images on both backbones, the loudest possible refusal, with zero uncertified automated decisions. The practical reading for a plant: the 50/50 split is not a requirement but a choice on a smooth budget curve, and whatever budget is spent, the certificate semantics do not degrade — only the coverage does.

5.6. How much calibration data buys a certificate: a planning curve

The budget sweep above answers what happens when calibration data is taken *away*. A plant scoping a deployment asks the inverse, and it is the more actionable question: given a target rate, how many labelled calibration images must be collected before the gate will certify at all? The floor $\alpha_{\text{min}} = 1/(n_{\text{cal}} + 1)$ inverts exactly, so this has a closed-form answer — certifying rate α on a category requires

$$n_{\text{cal}} \;\geq\; \lceil 1/\alpha \rceil - 1, \tag{9}$$

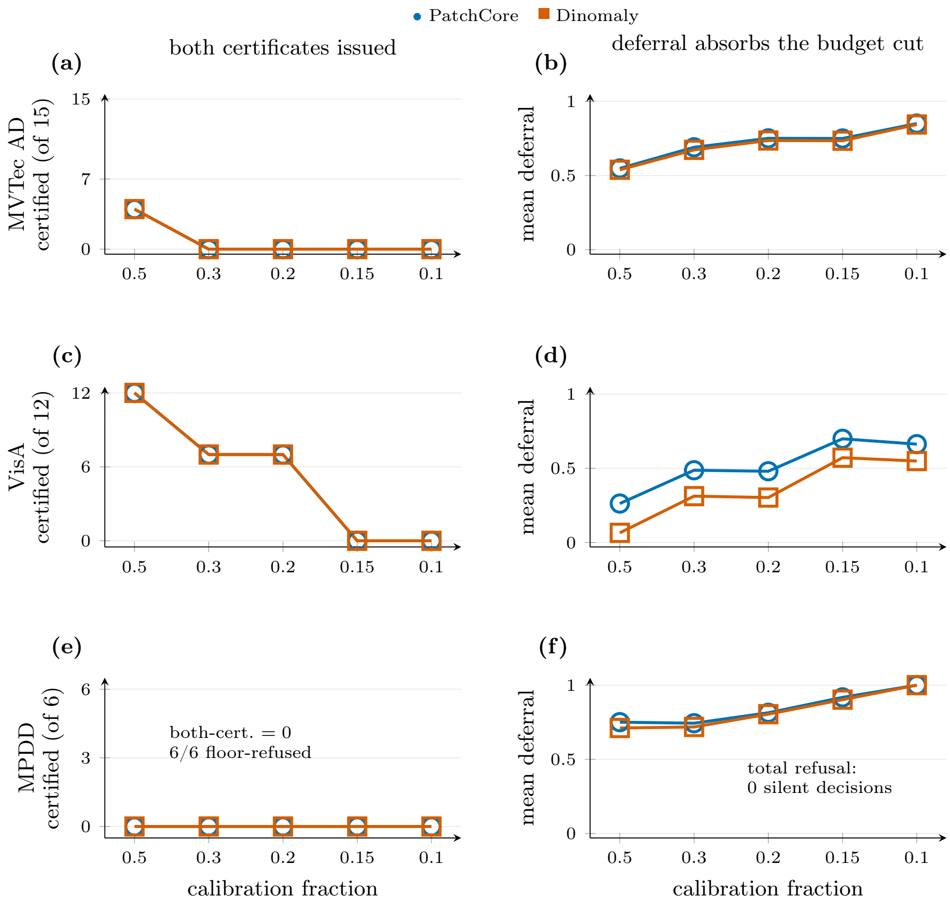

Figure 6: Calibration-budget sensitivity at the paper's operating point ($\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$; $R = 20$, five seeds; post-freeze exploratory). Rows are MVTec AD (a,b), VisA (c,d), and MPDD (e,f). Left panels show categories certified on both axes as the calibration fraction shrinks; right panels show mean deferral. Shaded bands span seed minima to maxima. Lost budget is absorbed by deferral, never by silent error: at fraction 0.10 MPDD refuses everything (deferral exactly 1.0, zero silent decisions). The flat zero in panel (e) for MPDD at fraction 0.10 is a measured refusal across all 20 repeats and five seeds — the floors refuse all six categories, so both-certified is identically zero and deferral saturates at 1.0 rather than crossing into silent mis-coverage.

of the relevant class (defective for escaped-defect, good for false-reject), per category, over and above whatever pool the detector itself was fitted on. Figure 7 turns Equation (9) into a procurement curve and prices the three benchmarks against it. The arithmetic is analytic, but we verify it against

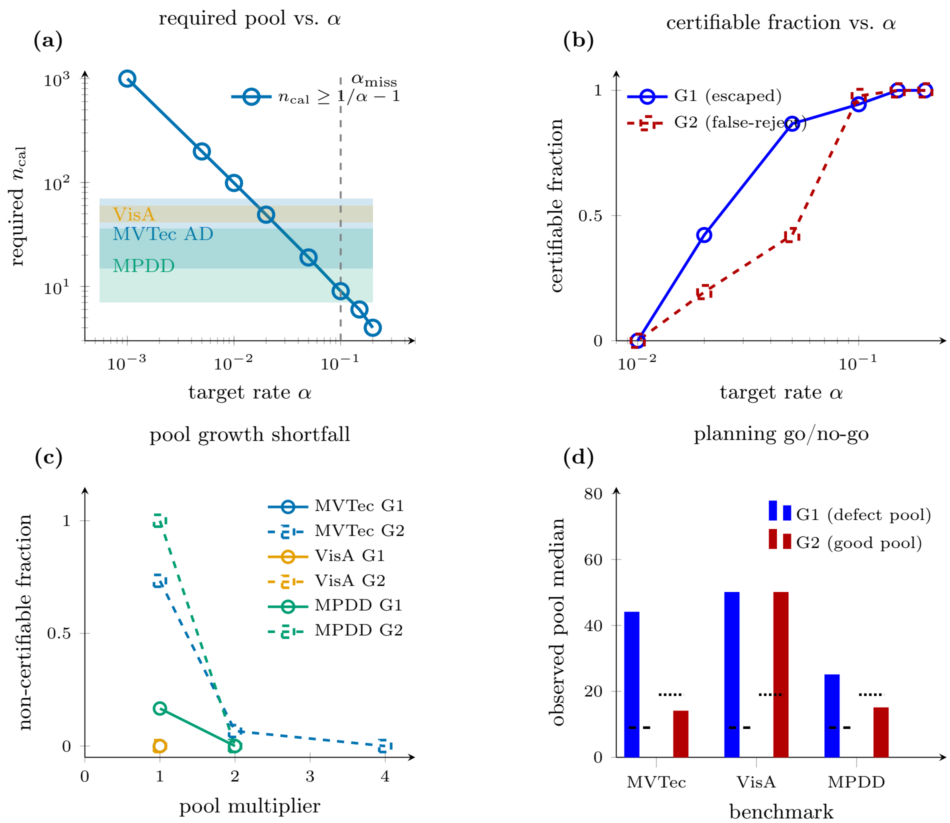

Figure 7: Calibration-pool planning curves from the frozen per-category pool sizes. (a) Required n_{cal} vs. target rate α (Equation (9)); shaded bands are observed defective-pool ranges, and the dashed line marks $\alpha_{\mathrm{miss}} = 0.10$. (b) Certifiable category fraction vs. α (G1 escaped-defect, solid; G2 false-reject, dashed), averaged over both backbones. (c) Pool-growth shortfall at the operating point as the *non*-certifiable fraction (zero = already fully certified). (d) Go/no-go check: bars are observed median pool sizes; dashed segments are the required n_{cal} on each axis.

the frozen artifacts: at the paper's operating point it reproduces every frozen certified count exactly (6/6 benchmark × backbone checks, MVTec AD G1 15/15 / G2 4/15, VisA 12/12 / 12/12, MPDD 5/6 / 0/6).

Three readings, and the first reframes the paper's own refusals. The rates certified here are *cheap*: $\alpha_{\mathrm{miss}} = 0.10$ costs 9 defective images per category and $\alpha_{\mathrm{fr}} = 0.05$ costs 19 good ones. Consequently the G2 refusals reported throughout — MPDD 0/6, MVTec AD 11/15 — are not a deep obstruction but a shortfall of roughly a factor of two in test-split allocation: holding the protocol fixed and only doubling each category's good pool takes MPDD from certifying nothing to certifying *all* six categories, and MVTec AD from 4/15 to 14/15 (Figure 7(b); 4× completes MVTec AD at 15/15). MPDD's per-category test-good pools are 26–32 images, about half of the 38 that a 50/50 split would need to clear 19. Second, the escaped-defect axis is already nearly paid for: MVTec AD and VisA clear the α_{miss} floor in every category as collected, and MPDD misses only connector, whose 14 test-defective images yield $n_{\text{cal}}^{\text{def}} = 7 < 9$. Third, the expense is concentrated entirely in the tight rates a safety-critical line would actually want: $\alpha = 0.01$ requires 99 per category and $\alpha = 0.001$ requires 999, which is 5–50× what any of these benchmarks supply. That is the quantitative form of the scope boundary we state in Section 7: on data of this scale the demonstrable object is the refusal discipline, and reaching a 1%-escape certificate is a data-collection problem rather than a methodological one. We state the planning rule in that form — Equation (9) is a hard procurement requirement, not a tuning knob, precisely because falling below it produces a refusal instead of a quietly weaker certificate.

5.7. Comparison with a conformal single-threshold baseline

The natural named baseline a reviewer would ask for is not a hand-picked straw man but the single-threshold construction our own G1 certificate specializes: split-conformal Conformal Risk Control (CRC) [36], applied with one per-category threshold at $\alpha_{\text{miss}} = 0.10$ and no defer option. For the 0/1 miss loss this threshold is exactly our G1 split-conformal threshold (an identity proved in the gate module's implementation), so CRC and our gate share the same finite-sample escaped-defect guarantee and certify the same cells; the only degree of freedom is what each does with the ambiguous middle. We recompute CRC on the identical canonical score dumps, under the identical protocol ($\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$, $R = 20$ repeated stratified 50/50 calibration/evaluation splits, five seeds, both backbones), so every cell in Table 3 is apples-to-apples with the main results. The pooled rate comparison uses Equation (10). CRC has no mechanism to certify the false-reject side at all — with one threshold, every non-passed image is an auto-reject, not a deferral — so its G2 column is structurally n/a.

$$
\begin{aligned}
\widehat{r}_{\text{miss}}(m) &= \frac{\sum_{i\in\mathcal{P}} \mathbf{1}\{y_i = \text{defective}\}\,\mathbf{1}\{D_m(s_i; c_i) = \text{pass}\}}{\sum_{i\in\mathcal{P}} \mathbf{1}\{y_i = \text{defective}\}},\\
\widehat{r}_{\text{fr}}(m) &= \frac{\sum_{i\in\mathcal{P}} \mathbf{1}\{y_i = \text{good}\}\,\mathbf{1}\{D_m(s_i; c_i) = \text{reject}\}}{\sum_{i\in\mathcal{P}} \mathbf{1}\{y_i = \text{good}\}},\\
\widehat{r}_{\text{def}}(m) &= \frac{1}{|\mathcal{P}|}\sum_{i\in\mathcal{P}} \mathbf{1}\{D_m(s_i; c_i) = \text{defer}\}.
\end{aligned}
\tag{10}
$$

Table 3: Our dual gate vs. a single-threshold CRC baseline controlling the same escaped-defect risk ($\alpha_{\text{miss}} = 0.10$), pooled over both backbones, five seeds, and all categories ($n_{\text{cells}} = 2 \times 5 \times n_{\text{categories}}$ per benchmark). CRC certifies the same escaped-defect cells (G1) but has no false-reject certificate (G2 n/a) and must auto-reject the entire ambiguous band instead of deferring it.

Benchmark	Method	G1 cert.	G2 cert.	Escaped	False-reject	Deferral
MPDD	Our dual gate	50/60	0/60	6.6%	0.0%	73.1%
MPDD	CRC (single thr.)	50/60	n/a	6.6%	36.9%	0.0%
VisA	Our dual gate	120/120	120/120	7.6%	3.0%	16.4%
VisA	CRC (single thr.)	120/120	n/a	9.5%	16.2%	0.0%
MVTec AD	Our dual gate	150/150	40/150	7.3%	0.5%	54.4%
MVTec AD	CRC (single thr.)	150/150	n/a	8.5%	3.1%	0.0%

On MPDD, where the false-reject floor refuses G2 entirely (Section 5.3), the contrast is starkest: CRC pays 36.9% false-reject to hold the same 6.6% escaped-defect rate our gate holds at 0.0% false-reject, by deferring 73.1% of images instead of auto-rejecting them. VisA and MVTec AD show the same direction at smaller magnitude (16.2% vs. 3.0% false-reject on VisA; 3.1% vs. 0.5% on MVTec AD) — CRC's realized escaped-defect rate is also somewhat higher on these two benchmarks (9.5% vs. 7.6% on VisA, 8.5% vs. 7.3% on MVTec AD), consistent with both methods controlling the same risk to within realized-rate sampling noise across repeats rather than producing bit-identical per-repeat routes. The comparison isolates exactly what the dual-gate design buys over the nearest published single-threshold alternative: not a better escaped-defect guarantee (both share the same conformal construction) but a false-reject guarantee CRC cannot offer, purchased by turning auto-rejects into deferrals. Figure 8 plots the three rates for both methods side by side across all three benchmarks, making the MPDD contrast in particular immediately visible. A per-backbone breakdown (Appendix Table A.3) confirms the pooled contrast is not carried by one backbone, and exposes that CRC's VisA false-reject is itself strongly backbone-dependent (28.8% PatchCore vs. 3.7% Dinomaly) — the dual gate's deferral mechanism matters most exactly where the detector underneath is weakest.

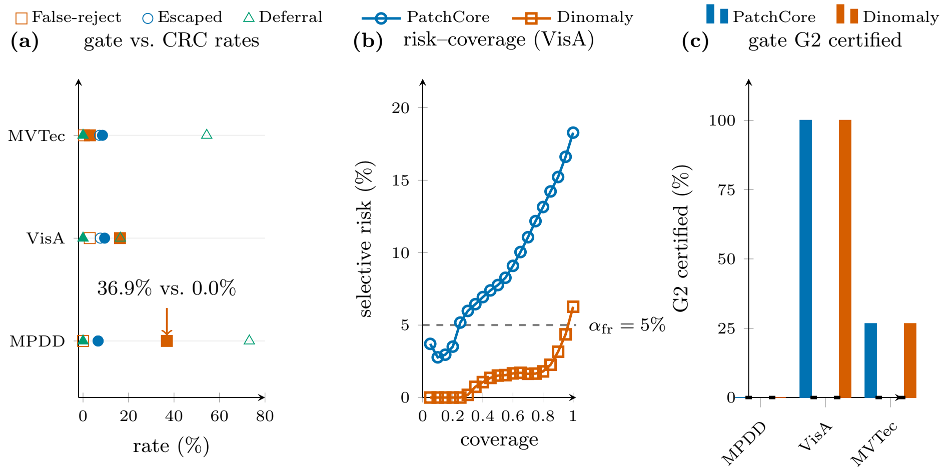

Figure 8: Dual gate versus single-threshold CRC baseline, pooled over both backbones and five seeds (Table 3). **a:** Escaped-defect, false-reject, and deferral rates side by side across all three benchmarks. CRC has no deferral mechanism, so every non-passed image becomes an auto-reject; on MPDD this drives its false-reject rate to 36.9% against 0.0% for our gate at the same escaped-defect rate. **b:** Selective-prediction risk-at-coverage curves per backbone on VisA — the CRC baseline specializes to per-category selective classification with confidence-based abstention, but no coverage level reaches the 5% false-reject guarantee the gate certifies. **c:** Per-backbone G2 certification: the gate certifies both backbones on VisA (60/60 cells each), one on MVTec AD (20/75 each), and neither on MPDD (0/30 each); CRC certifies zero cells on every benchmark and backbone (0/75 MVTec AD, 0/60 VisA, 0/30 MPDD, per backbone) — it cannot deliver the false-reject guarantee at any coverage.

As a second, non-conformal reference line we also compute the field-standard selective-prediction risk-coverage curve [31] (per-category best-F1 threshold, margin-based abstention) on the same score dumps. Its area-under-the-risk-coverage-curve (AURC, lower is better) ranges from 0.0019 (MVTec AD, Dinomaly) to 0.0849 (VisA, PatchCore); at $\sim 80\%$ coverage the realized risk is 0.0625/0.0154 (MPDD, PatchCore/Dinomaly), 0.1315/0.0180 (VisA, PatchCore/Dinomaly), and 0.0095/0.0026 (MVTec AD, PatchCore/Dinomaly). This is a fundamentally different guarantee — a curve traced out by sweeping coverage post hoc, not a fixed pre-specified certificate at a chosen risk level — so we report it as context rather than a like-for-like comparison to Table 3.

5.8. Deployment cost analysis

Table 3 certifies the dual gate's rates but not their economic consequence: every deferral is a decision routed to a human reviewer, and every false-reject the gate avoids is one the CRC baseline pays for instead. We translate both operating points into practitioner economics under three ASSUMED, explicitly labeled illustrative deployment scenarios spanning a 30× throughput range — a low-volume precision line (100 parts/hour, $5,000 per escaped defect, \$150 per false reject, a \$45/hour reviewer at 120 parts/hour), a mid-volume general industrial line (600/hour, \$300/\$15, a \$30/hour reviewer at 240/hour), and a high-volume commodity line (3,000/hour, \$25/\$2, a \$26/hour reviewer at 450/hour) — applied to the certified rates of Table 3 (neither the rates nor the frozen calibration are altered; only the dollar figures are scenario-dependent). The full nine-cell benchmark × scenario grid and script are in the released artifact set of the code-and-data deposit; Table 4 shows the mid-volume scenario.

Table 4: Deployment cost analysis, mid-volume general industrial scenario (600 parts/hour, \$300 per escaped defect, \$15 per false reject, \$0.125/part review labor). Reviewer load is throughput × deferral rate; CRC's is 0 by construction (no deferral mechanism — every non-passed item is an automatic reject, not a human decision). Breakeven review cost/part is the human-review unit cost above which CRC would become cheaper than the dual gate, holding this scenario's other costs fixed; headroom is that breakeven divided by the \$0.125/part assumed here. The all-human row is a labor-only comparator (\$30/hour at 240 parts/hour); no human miss/false-reject model is available, so its \$75/hour is not a like-for-like total-error-cost estimate.

Benchmark / policy	Reviewer load	Gate/policy \$/hr	CRC \$/hr	Gate savings	Headroom
MPDD	438.5/hr (1.83 FTE)	\$11,990	\$15,252	+\$3,262/hr	61×
VisA	98.4/hr (0.41 FTE)	\$14,024	\$18,547	+\$4,522/hr	369×
MVTec AD	326.1/hr (1.36 FTE)	\$13,134	\$15,661	+\$2,527/hr	63×
All-human baseline	600.0/hr (2.50 FTE)	\$75 labor only	—	—	—

The all-human row exposes the labor floor behind the scenario: reviewing all 600 parts/hour at 240 parts/hour per reviewer requires 2.50 FTE and costs \$75/hour. It is deliberately labor-only because the frozen artifacts contain no human miss or false-reject model; it therefore contextualizes staffing, not comparative decision quality.

The dual gate is cheaper than CRC in all nine benchmark × scenario combinations we compute, though the economic case varies by deferral cost. On VisA and MVTec AD, where deferral rates are moderate (16.4% and 54.4% respectively), the gate saves \$2,527–\$4,522/hour at mid-volume with substantial headroom (63–369×). MPDD is economically viable only in scenarios where high review capacity is available or tolerable: at mid-volume its 73.1% deferral translates to 438.5 parts/hour (1.83 FTE), saving \$3,262/hour against CRCś 36.9% false-reject rate, but at high-volume the same deferral becomes 2,192.7 parts/hour (nearly five FTEs) — positioning MPDD as a high-review diagnostic scenario rather than a universally economical deployment. The saving is not an artifact of favorable escaped-defect rates: on MPDD the two methods share the identical escaped-defect rate by the proven G1≡CRC threshold identity, and on VisA/MVTec AD the gate's escaped rate is additionally somewhat *lower* than CRC's. The least robust cell is MVTec AD under the high-volume commodity scenario: human review would need to cost more than \$0.69/part before CRC becomes cheaper there, about 12× the \$0.058/part assumed in that scenario. Every other cell has 17–1,833× headroom. This model omits reviewer error rate, queueing dynamics, and an appeal channel for CRC's false-rejected good parts; modeling the latter would raise CRC's cost further, so the comparison is if anything conservative toward CRC. Its dollar figures are scenario-dependent by construction, while the throughput-invariant breakeven quantities above are the assumption-robust takeaway.

Figure 9 shows the deployment economics of the certified gate versus the CRC baseline.

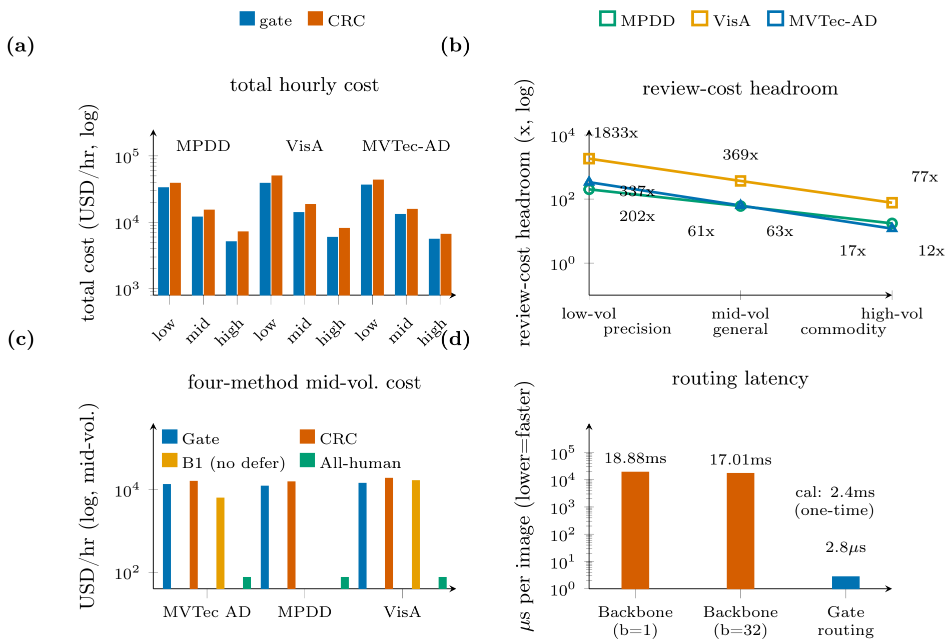

Figure 9: Deployment economics of the certified gate versus the CRC single-threshold baseline at frozen rates. Scenario groups: low-volume precision (100 parts/hour), mid-volume general (600), high-volume commodity (3,000); scenarios are assumed illustrative inputs. **a:** Total hourly cost (gate vs. CRC) across all nine benchmark–scenario cells: the gate is cheaper in every cell. **b:** Review-cost headroom — the multiple by which the assumed review cost can rise before the gate stops being cheaper: $12\times$–$1833\times$ (log; break-even at $1\times$). **c:** Four-method total hourly cost at mid-volume general, including the original B1 binding-demo arm (no deferral, post-hoc, unavailable for MPDD); the gate is cheapest on MVTec AD and VisA. **d:** Routing latency: the gate adds $2.8\,\mu$s per image amortised (right bar), against 18.9 ms backbone inference at batch size 1 and 17.0 ms batched (left bars). Calibration is a one-time 2.4 ms per reset.

5.9. Latency and deployment cost

The gate is a thin decision layer on top of whatever detector already produces the anomaly score, so the only cost it adds is the conformal calibration (once, per calibration refresh) and the per-image routing comparison. We measured both on this work's hardware, and we place them alongside the backbone forward pass that dominates the end-to-end budget (Table 5). Routing an image is an $O(1)$ threshold comparison inside an $O(\text{sort})$ calibration; the measured per-image routing cost is $2.8\,\mu$s on CPU — roughly four orders of magnitude below the backbone forward pass and under 0.02% of Dinomaly's per-image GPU inference. Calibrating the full three-way gate over one 861-image calibration half across all 15 MVTec strata (the $\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$ primary configuration) takes 2.4 ms end to end, and is amortized over an entire evaluation run rather than paid per image. The practical conclusion is that certification is free relative to detection: a deployment already able to afford the detector can add the escaped-defect / false-reject certificate at no measurable latency cost. The backbone rows are the honest budget the gate sits on — Dinomaly measured here on an RTX 5090 Laptop GPU, PatchCore's row taken from the authors' reported figure (not measured here, and both hardware- and coreset-dependent). We flag the substrate mix these numbers span, so no cross-row comparison is over-read: the gate rows are timed on CPU over the MVTec calibration scores (861-image cal half, 15 strata), the Dinomaly backbone row on GPU over 250 real VisA test images (the VisA branch-A seed-0 checkpoint), and the PatchCore row is the authors' published ∼1%-coreset figure whereas our scoring uses the 10%-coreset configuration — so the table brackets the order of magnitude of the backbone budget the gate rides on, and it is the μs-scale CPU gate overhead, not any CPU-vs-GPU or MVTec-vs-VisA backbone comparison, that the paper contributes.

Table 5: Per-image latency and deployment cost. The two gate rows and the Dinomaly backbone row are *measured on this work's hardware* (gate: Intel Core Ultra 9 275HX CPU, 24 threads; Dinomaly: NVIDIA RTX 5090 Laptop GPU, torch 2.12.0, batch 1, median over 250 real VisA test images, full forward + anomaly-map + image-score pass). The PatchCore backbone row is *reported by the authors* (not measured here; Roth et al. [2], ∼1%-coreset configuration on their hardware) and is included only to bracket the backbone budget — backbone timing is hardware- and configuration-dependent, whereas the gate overhead below it is the measured constant this paper contributes. Sources: the frozen latency measurement dumps.

Component	Per-image cost	Notes
Backbone forward pass (dominates the budget)		
Dinomaly (DINOv2-b, 148M) — measured, GPU	18.9 ms (p95 23.1)	53 img/s at batch 1; 17.0 ms/img batched (B=32)
PatchCore (WRN50, coreset) — *authors' figure*	$\lesssim$ 200 ms	Roth et al. [2], their hardware; not measured here
Gate overhead added on top (this paper) — measured, CPU		
Routing (per image)	2.8 μs	$O(1)$ compare; < 0.02% of the Dinomaly forward pass
Calibration (one-time)	2.4 ms (p95 2.8)	full 3-way gate, 861-img cal half, 15 strata; per refresh

5.10. G2 train-holdout promotion on MVTec AD (frozen remedy arm)

The preregistered calibration-efficiency remedy — recalibrating G2 on held-out *train*-good scores, gated behind a per-category Kolmogorov–Smirnov

(KS) exchangeability test with a mandatory audited-not-certified fallback — was computed after the freeze, for PatchCore only: post-freeze MVTec/VisA Dinomaly train-score dumps now exist, but they are checkpoint-dependent and do not provide the independent held-out pool required for a fresh G2 promotion calibration. The remedy arm therefore remains single-backbone by evidence eligibility, disclosed here rather than implied otherwise. It behaves as preregistered: G2-certifiable categories rise from 4/15 to 13/15 in seeds 0–3 and 12/15 in seed 4. The KS gate does real work rather than rubber-stamping: leather fails it in all five seeds (its train-good and test-good score distributions are not exchangeable) and screw additionally fails in seed 4 — each is reported audited-not-certified, never silently promoted, and the seed-4 screw case shows a category certified under the primary protocol can still be *refused* under the holdout arm when exchangeability breaks. Toothbrush remains uncertifiable at $\alpha_{\mathrm{fr}} = 0.05$ under both arms, exactly as preregistered (n^{good} too small in either pool). K1 is not tripped in any promoted seed aggregate.

What the remedy costs and delivers (post-freeze pricing).. Figure 10 shows the per-category certification flip.

Pricing the flip on the frozen post-hoc remedy-pricing artifacts qualifies it in both directions. On the cost side, certification is *not* bought with more deferral: opening the auto-reject region (a finite t_{hi} where the primary protocol refused) *reduces* deferral in every rescued category — capsule certifies at 3.9% mean deferral (from 75.1% under refusal), bottle 6.1%, tile 7.5%, metal_nut 7.7%; the most expensive rescue is carpet at 45.2%. On the honesty side, two flags travel with the headline. First, the already-certified screw *degrades* under the promotion arm: its realized false-reject mean rises to 8.8% (range 4.8–12.1% across seeds, above the 5% target in four of five seeds and above the +3 pp tier-1 tolerance in three) before it KS-fails outright in seed 4 — the 13/15 figure is therefore strictly a seeds-0–3 statement, and the promotion arm is a per-category choice, not a global upgrade. Second, grid, zipper, and pill realize false-reject marginally above the 5% point target in some seeds (max 6.8%/5.6%/7.7% respectively) though always inside the tier-1 tolerance. G1 is structurally unaffected by the arm; every rescued category's realized escaped-defect rate stays within target plus tolerance in every seed.

5.11. MPDD train-holdout rescue (post-freeze exploratory)

We report the primary-protocol 0/6 of the preceding section as MPDD's thesis-confirming headline; the train-holdout arm below is a clearly secondary, post-freeze, PatchCore-only result. The train-holdout calibration-efficiency lever the frozen MVTec remedy exercises (Section 5.10) turns MPDD's 0/6 primary refusal into a 4/6 rescue, a vivid single-benchmark illustration of the same lever. MPDD's train-good pools are large (54–289 per category)

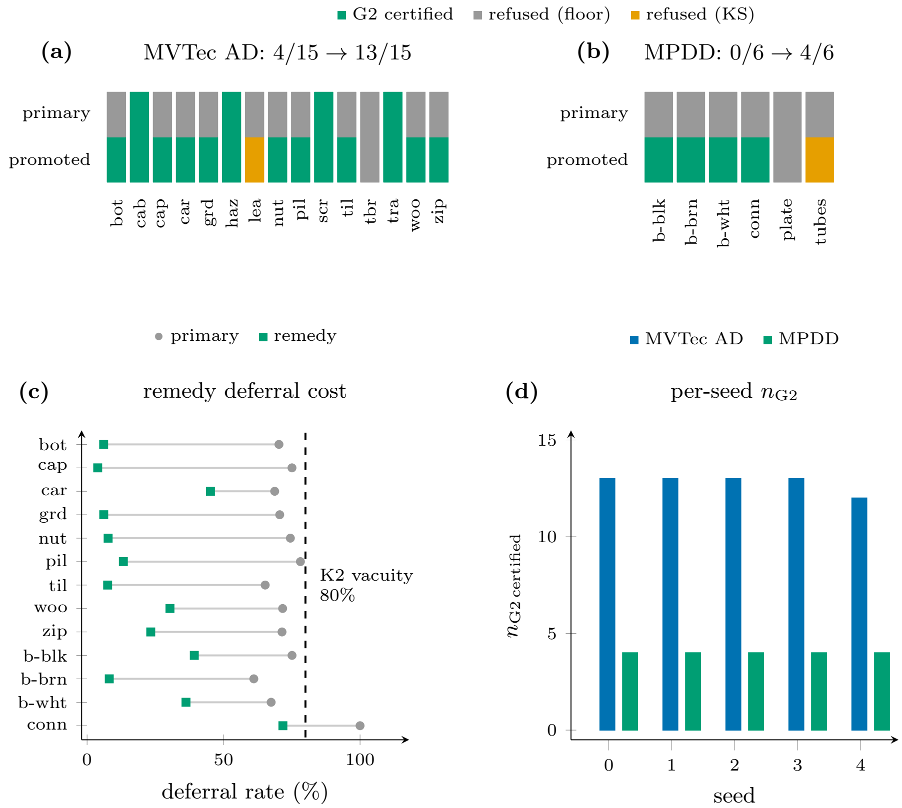

Figure 10: The G2 train-holdout remedy (PatchCore; seeds 0–3 identical, seed 4 differs on screw). **a:** MVTec AD: 4/15 → 13/15, leather KS-refused every seed. **b:** MPDD: 0/6 → 4/6. Category codes — MVTec AD: bot/cab/cap/car/grd/haz/lea/nut/pil/scr/til/tbr/tra/woo/zip; MPDD: b-blk/b-brn/b-wht/conn/plate/tubes. Tiles: certified (green), refused at floor (grey), refused by KS (amber). **c:** Remedy deferral cost per rescued category: promotion opens the auto-reject region and *reduces* deferral for most categories; connector (MPDD) remains near-vacuous at 71.7%. **d:** Per-seed $n_{\mathrm{G2\,certified}}$ under the promoted arm: 13/15 stable in seeds 0–3 (screw degrades to 12/15 in seed 4 when KS fails); MPDD holdout certifies 4/6 in all five seeds.

where its test-good pools are tiny, so recalibrating G2 on a 20% held-out train-good pool (the flag-gated train-holdout calibration mode, 20% holdout fraction; PatchCore only, as on MVTec — MPDD has no Dinomaly train-score dump, while the later MVTec/VisA dumps are checkpoint-dependent and not eligible independent holdouts for this arm) clears the false-reject floor for the 5/6 categories whose holdout pool reaches $n \geq 19$. The same per-category KS exchangeability gate then adjudicates each promotion, and it does decisive work: G2-certifiable rises from 0/6 to 4/6, identical in all five seeds. Two categories are honestly held back — metal plate, whose holdout pool (11) is still below the floor, and tubes, which fails the KS gate (Benjamini–Hochberg (BH)-adjusted $p \approx 1.2 \times 10^{-5}$; its train-good and test-good score distributions are demonstrably non-exchangeable) and is reported audited-not-certified rather than promoted. The remaining four (bracket black, bracket brown, bracket white, connector) promote cleanly (KS $p_{\mathrm{BH}} \geq 0.44$ every seed). Connector is the one category G1 refuses under the primary protocol ($n_{\mathrm{cal}}^{\mathrm{def}} = 7 < 9$ defective) yet the arm G2-certifies (holdout-good $26 \geq 19$): the two certificates gate on independent sample axes — defect count for escaped-defect, good count for false-reject — so this is coherent, not contradictory. Even certified, connector still defers 71.7% of all images under the arm (down from 100% under refusal; G1's empty auto-pass region caps what G2 alone can recover), so that certificate remains operationally near-vacuous. The cheapest rescue is bracket brown at 8.1% mean deferral with realized false-reject 0.0% in every seed; the one realized-rate flag is bracket black, whose per-seed false-reject hits 10.0% in two of five seeds (mean 5.1%) — the certificate bounds the *expected* rate at 5%, but the realized rate is seed-unstable there, and a deployment should treat the category as marginal. So the count-only floor of 5/6 is audited down to a genuinely certified 4/6 — the KS guardrail keeping the rescue honest, exactly the exchangeability risk MPDD's non-homogeneous backgrounds were expected to pose. As a sanity check (no gate), the holdout arm's 80%-memory-bank test scores rank-correlate with the primary 100%-bank scores at Spearman $\rho \geq 0.918$ across all 6×5 category-seed cells (max $|\Delta\text{image-AUROC}| = 0.040$), confirming the same model was scored. This arm is post-freeze exploratory and flag-gated prereg-neutral (it re-freezes nothing and is never in the confirmatory Holm family); the $0/6 \rightarrow 4/6$ contrast on the same benchmark and the same scores is a vivid single-benchmark demonstration of the calibration-efficiency lever. The larger, *confirmatory* instance of the same lever is the frozen MVTec train-holdout remedy (Section 5.10), which lifts G2 from 4/15 to 12–13/15 across seeds (13 in seeds 0–3, 12 in seed 4). The primary-protocol 0/6 stays the honest headline for this industrial benchmark: on real metal parts the gate certifies nothing at the 5% false-reject budget because the data cannot pay for it — exactly what a refusal-first design should do.

5.12. When a fixed threshold over-promises (post-hoc binding demonstration)

The excess-AURC audit of Section 5.14 shows practice earns deferral skill, but it does not directly exhibit the sharper event: a category where a *naive fixed threshold* actually violates a target error rate while the certified gate holds. We test that event directly, post-hoc. For each (backbone, category) we take the naive practitioner's operating point — a single global best-F1 threshold (B1) applied to every eval image with no deferral — and compute its realized escaped-defect and false-reject rates on the held-out eval half; alongside it we compute the certified gate's realized rates and its deferral cost on the same split (mean over the $R = 20$ repeats, at backbone seed 0). A cell *binds* when the fixed threshold's realized rate exceeds the target ($\alpha_{\text{miss}} = 0.10$ or $\alpha_{\text{fr}} = 0.05$) while the gate's stays within target+3pp.

Such cells exist, on both benchmarks, both axes, and both backbones. Restricting to the strongest evidence — cells where the gate is certified on that axis in all 20 repeats and the bind holds in all five backbone seeds — there are 27: on MVTec AD, 3 escaped-axis (Dinomaly on capsule, pill, screw) and 3 false-reject-axis (PatchCore on screw; Dinomaly on cable, transistor); on VisA, 4 escaped-axis and 17 false-reject-axis cells. Crucially the effect is not an artifact of the gate abstaining on everything: the cleanest cases pay a small deferral cost. On MVTec AD a global threshold lets 24.2% of defective screws escape; the certified gate, deferring just 4% of images, holds the escaped rate at 8.0% (under target) because its per-category (Mondrian) calibration puts the auto-pass boundary where a global threshold cannot. On the false-reject axis, the same global threshold auto-rejects 19.5% of good screws while the gate holds false-reject at 5.0% (certified, 14% deferral); on VisA the extreme case is capsules, where a global threshold would auto-reject 96.8% of good parts and the gate holds false-reject at 3.7%. The mechanism the demonstration exposes is that a single fixed *global* threshold cannot serve categories with different score scales at once, so it over-promises on whichever it mis-fits; the certified gate refuses (defers) exactly where it cannot honor the target and calibrates per category elsewhere. This is a post-hoc, exploratory illustration — not a preregistered arm — and is reported verdict-honestly: had no cell bound, we would have kept the "not demonstrated" framing. Figure 11 shows every escaped-defect-axis binding cell: in each row the naive threshold sits beyond the target line while the certified gate holds it, at the annotated deferral price.

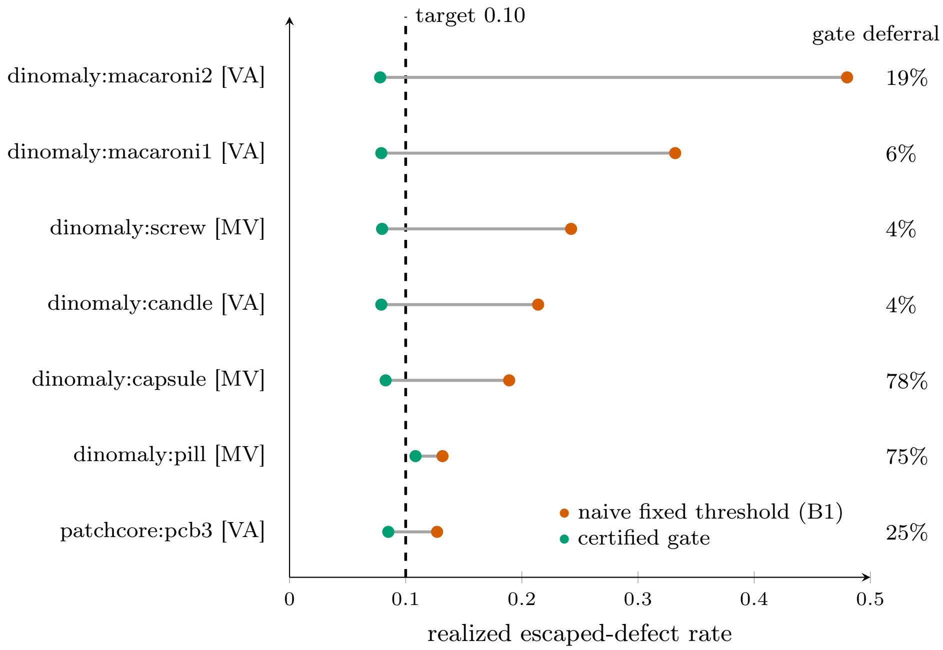

Figure 11: Binding cells on the escaped-defect axis (post-hoc). Per (backbone, category) cell: realized escaped-defect rate of the naive global best-F1 threshold (vermillion) vs. the certified gate (bluish green), with the gate's deferral annotated at right; the dashed line is the $\alpha_{\text{miss}} = 0.10$ target. MV = MVTec AD, VA = VisA.

Figure 12 reports the corresponding false-reject binding cells under the same encoding.

5.13. Validity audit V1 (tier-1) and the kill-gates

Per frozen amendment A3, tier-1 is reported *per axis*, with the false-reject axis evaluated only over G2-certified cells: wherever G2 is refused the auto-reject region is empty and the measured false-reject rate is identically 0 — a pass without evidence that A3 excludes from every count. On the escaped-defect axis, tier-1 passes in all 150 MVTec cells (2 backbones $\times 5$ seeds $\times 15$ categories), all 120 VisA cells ($2 \times 5 \times 12$), and all 60 MPDD cells ($2 \times 5 \times 6$), every rate within target+3pp. On the false-reject axis, MVTec has 40 evaluable cells (4 G2-certified categories $\times 2 \times 5$) and passes 40/40, with the other 110 cells reported G2-REFUSED and excluded; on VisA every category is G2-certified, so all 120/120 cells are evaluable and pass non-vacuously; on MPDD every category is G2-refused under the primary protocol, so all 60 false-reject cells are G2-REFUSED and excluded (the mirror image of VisA, and the primary-protocol counterpart to the train-holdout rescue of Section 5.11). Tier-1 is a necessary-not-sufficient pipeline check — a conservative gate realizes rates well below target, so passing validates the calibrate/route/evaluate machinery rather than demonstrating sharpness. Consequently K1 is not tripped (the A3 exclusion cannot change

Figure 12: Binding cells on the false-reject axis (post-hoc). Same encoding as Figure 11; the dashed line is the $\alpha_{\text{fr}} = 0.05$ target. The VisA PatchCore capsules cell (top) is the extreme case: a global threshold auto-rejects 96.8% of good parts where the certified gate holds 3.7%.

K1's count: excluded cells cannot fail) (0 violations in every (backbone, seed) aggregate, against the 5-cell MVTec/VisA kill threshold and, on MPDD, against a proportionally-scaled 2-cell threshold[2]) and K2 is not tripped (on MVTec/VisA at most 1 vacuous category — VisA/PatchCore/seed 2 — against the 8-category/0.80-deferral threshold; on MPDD at most 2 vacuous categories per aggregate against the scaled 4-category threshold), in all 30 aggregates across the three benchmarks (Table 6). Table A.2 reports seed-0 median per-category deferral, its mean where computed, and the extreme categories, for each benchmark and backbone. MVTec AD's deferral is elevated because G2 refusal empties the auto-reject band in 11/15 categories, though still far from the vacuity threshold. VisA's deferral drops sharply because G2 certifies every category there, so the auto-reject band is live throughout — direct operational evidence that the elevated MVTec deferral is the price of honest G2 refusal, not a property of the gate. MPDD sits at the far end: G2 is refused in all 6 categories under the primary protocol, exactly what 0/6 certification predicts. These are three honest operating points, not a defect and a fix: the elevated MVTec and MPDD deferral is the cost of refusing to certify small good-calibration pools, and VisA's low deferral is what the same gate, same code path, spends when the good pools are large enough to certify every category.

The per-repeat validity audit is shown in Figure 13.

Figure 13 opens up the tier-1 aggregates above to the per-repeat level: the certified clouds sit at or below target on the overwhelming majority of repeats, the per-(benchmark, backbone) means sit below target everywhere, and the visible excursions are single repeats on small cells — the fluctuation the marginal certificate semantics predict, not a validity breach (Section 5.5 shows the same invariance holds down to a tenth of the calibration budget).

Figure 14 shows median deferral per category for both backbones.

[2]Absolute kill thresholds are un-trippable at $n_{\mathrm{cat}} = 6$, so MPDD uses proportional scales (K1$\rightarrow$ 2, K2$\rightarrow$ 4); diagnostic only.

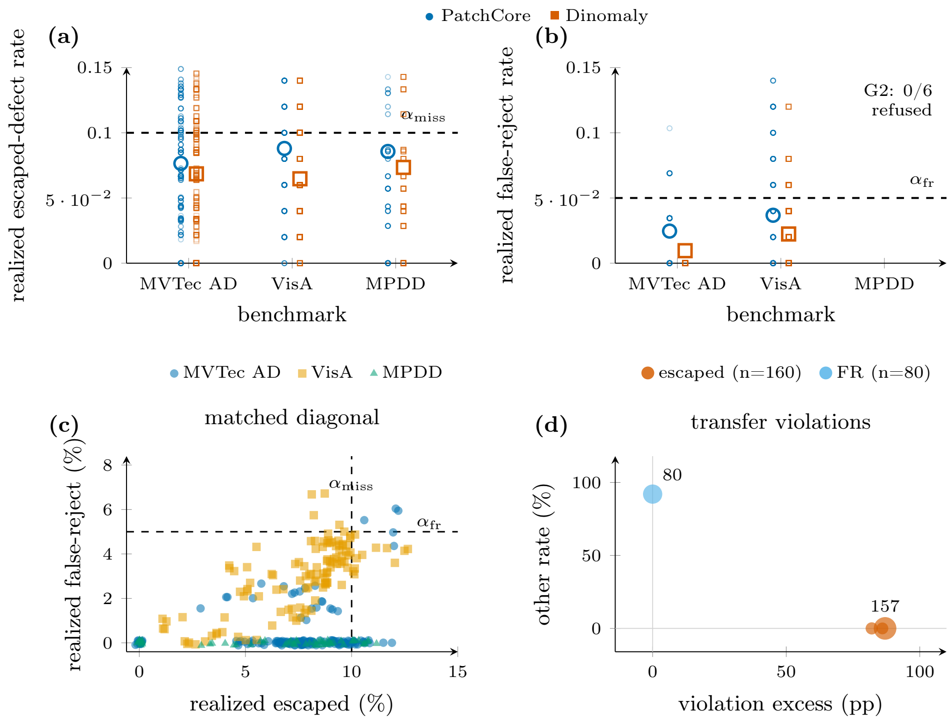

Figure 13: Per-repeat realized rates (frozen protocol; 6,400 escaped-defect and 3,200 false-reject certified-cell measurements). **a:** Escaped-defect rate vs. $\alpha_{\text{miss}} = 0.10$; **b:** false-reject rate vs. $\alpha_{\text{fr}} = 0.05$ (MPDD has no G2-certified cells under the primary protocol). Heavy dashes are per-(benchmark, backbone) means; every mean sits below target. **c:** Matched-diagonal validity cloud (330 cells, same-detector diagonal from the cross-detector transfer experiment): the certified region sits at or below both targets, confirming the guarantee holds on the same-detector reference. **d:** Transfer violation cloud: 160 escaped-defect violations (transfer adds 82–87 pp above α_{miss}) and 80 false-reject violations; the two violation types separate cleanly by direction.

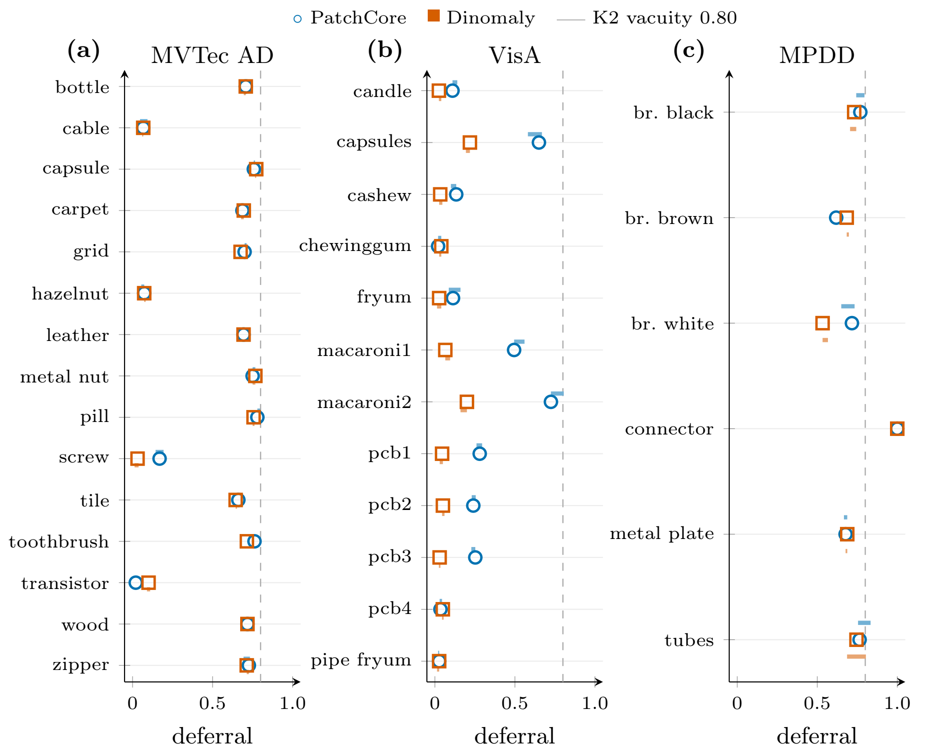

Figure 14: Median deferral per category, seeds 0–4 (dot = seed 0, shaded band = seed 0–4 min–max; $R = 20$), both backbones. **a:** MVTec AD. **b:** VisA. **c:** MPDD. Deferral is strongly category-heterogeneous and tracks certifiability (refused strata defer by construction). The K2 vacuity threshold (0.80, dashed) is approached only on a handful of G2-refused categories; seed spread is narrow except there.

Table 6: V1 tier-1 (per axis, per frozen amendment A3) and kill-gates, aggregated over the 30 (backbone, seed) cells across all three benchmarks. †False-reject evaluated only over the 4 G2-certified MVTec categories; the other 110 cells are G2-REFUSED and excluded from the count, never counted as (vacuous) passes. ‡On MPDD all 60 false-reject cells are G2-REFUSED under the primary protocol and excluded (see Section 5.11 for the train-holdout rescue). MPDD's K1/K2 thresholds are proportionally scaled to $n_{\mathrm{cat}} = 6$ (2 and 4). "—" marks a cell that does not apply to that row (the raw evaluable-cell counts above have no separate threshold or trip verdict; K5 never fires under the frozen design, so it reports no single value). Sources: the frozen MVTec AD, VisA, and MPDD analysis summaries and their per-cell validity dumps; frozen preregistration §9 amendment A3.

Axis	Value	Threshold	Tripped?
V1 tier-1 escaped-defect cells (MVTec AD)	150/150	—	—
V1 tier-1 false-reject cells (MVTec AD)†	40/40	—	—
V1 tier-1 escaped-defect cells (VisA)	120/120	—	—
V1 tier-1 false-reject cells (VisA, all G2-certified)	120/120	—	—
V1 tier-1 escaped-defect cells (MPDD)	60/60	—	—
V1 tier-1 false-reject cells (MPDD)‡	0 evaluable	—	—
K1 coverage violations (per cell)	0	≥ 5 (≥ 2 MPDD)	No (all 30)
K2 vacuous categories (per cell)	0 (1 in VisA/ PatchCore/seed 2; ≤ 2 on MPDD)	≥ 8 (≥ 4 MPDD)	No (all 30)
K3 backbone floor (mean AUROC)	≥ 0.982 (MVTec) / ≥ 0.903 (VisA) / ≥ 0.948 (MPDD)	< 0.90	No
K5 calibration floor	—	$\alpha_{\min} > 0.10$ (≥ 3 cat)	Cannot fire

5.14. Exploratory excess-AURC audit

As an exploratory (not confirmatory) readout scoped to one (backbone, seed, category) family of size 2 (practices B1 and B2; B3 could not run — see below), on the repeat-0 split at seed 0: on MVTec AD, PatchCore rejects the random-deferral null in 10/30 practice-cells after within-cell Holm, with 6/30 degenerate (per-category AUROC $= 1.0 \Rightarrow$ zero measurable headroom); Dinomaly rejects in 0/30 with 14/30 degenerate — an expected ceiling effect, since Dinomaly sits at per-category AUROC 1.0 in 8/15 categories on this split, leaving threshold practices no room to show skill above a near-perfect baseline. VisA resolves this ceiling effect: on the same seed-0 readout, PatchCore rejects in 14/24 cells and Dinomaly in 16/24, with 0 degenerate cells for either backbone — on a benchmark where the backbones are not saturated, field-standard threshold practices demonstrably do earn deferral skill over honest random deferral, so the audit's verdict axis is live rather than vacuously constructive. Across all five seeds the picture is stable (MVTec: PatchCore 48/150, Dinomaly 4/150; VisA: PatchCore 64/120, Dinomaly 42/120 practice-cell rejections). One comparability caveat: on VisA, Dinomaly is a multi-class model (one model per seed for all 12 categories) while PatchCore keeps per-category memory banks, so cross-backbone audit comparisons there are not architecture-controlled and are read per backbone, not head to head.

5.15. Confirmatory audit verdict (C2)

The exploratory readout above pools nothing; its higher-powered confirmatory counterpart pools the 15 MVTec categories per (practice, backbone). The confirmatory family is the four tests $\{\text{B1 fixed}, \text{B2 tuned}\} \times \{\text{PatchCore}, \text{Dinomaly}\}$; the train-good quantile practice B3 drops out because the frozen confirmatory pass loaded no train-good scores, the degradation from six to four tests preregistered in PREREG §4. B3 is not infeasible in general: a held-out train-good score pool does exist for PatchCore, so its B3 arm is completed post-hoc below. Post-freeze MVTec/VisA Dinomaly train-score dumps support a checkpoint-dependent practice sensitivity, but not an independent holdout eligible to extend the frozen confirmatory family; MPDD has no Dinomaly train-score dump. The full six-member family therefore remains unavailable. For each (practice, backbone) we pool the 15 categories into one evaluation half (repeat-0 split), fit B1 on the pooled calibration half and B2 per category, and test the excess area under the risk–coverage curve against the closed-form random-deferral null with a matched-abstention permutation test ($n_{\text{perm}} = 2000$, strata = category) and a category-blocked bootstrap CI. Every one of the four tests rejects the random-deferral null at the permutation floor $p = 1/2001 = 5 \times 10^{-4}$ (Holm-adjusted $p = 2 \times 10^{-3}$) in all five backbone seeds, with pooled excess-AURC in $[0.023, 0.050]$ and every bootstrap CI excluding zero (Table 8). The frozen confirmatory construction is the per-seed four-member Holm family, and it rejects in every one of the five seeds. The preregistration is silent on how to combine seeds, so we report the seed dimension as a robustness check rather than a family axis: the single cross-seed rollup (the seed-max of the per-seed p-values) was authored post-freeze as a one-shot convenience and is moot here, the verdict being identical under seed-min, seed-median, and seed-max. The constructive arm of C2 publishes: field-standard threshold practice earns real deferral skill over honest random deferral. Pooling recovers across the 15 categories the power that per-category saturation denies the exploratory single-category readout. Because the family rests on the repeat-0 split, we re-ran the byte-identical frozen construction on all 20 frozen splits post-freeze (exploratory; repeat-0 reproduced exactly): the verdict is stable in 20/20 repeats with zero flips — every (repeat, seed) cell rejects all four members after Holm, the worst raw p anywhere being 0.014 — so the split choice is not load-bearing.

B3 (train-good quantile), the family's feasible half, post-hoc.. Because a held-out train-good score pool exists for PatchCore (the holdout run scored the per-category train-good images alongside the test set), we complete the third practice for that backbone post-hoc, on the same pooled-category construction. B3 is constructive and consistent with B1/B2: across all five seeds it rejects the random-deferral null at the permutation floor $p = 5 \times 10^{-4}$, with pooled excess-AURC in $[0.053, 0.056]$ (larger than B1's and B2's) and every category-blocked bootstrap CI excluding zero; a cross-run sensitivity arm agrees. Added to the frozen four-member per-seed family as a fifth member, all five reject in all five seeds (Holm $p = 2.5 \times 10^{-3}$). This is labelled post-hoc and exploratory: the confirmatory verdict remains the frozen four-member family above. B3-Dinomaly is not added: the available MVTec/VisA train dumps are checkpoint-dependent rather than independently held out, and MPDD has no corresponding dump.

Table 7: Post-freeze train-side score availability. A cell is one (category, seed) score pool; n is the number of train-side or held-out-good scores in that cell. The Dinomaly rows are checkpoint-dependent practice sensitivities, not fresh G2 calibration pools or members of the confirmatory B3 family. The VisA PatchCore row is an independent holdout because those images were excluded from the memory bank.

Benchmark	Backbone / pool	Cells	n per cell (mean)	Evidence use
MVTec AD	Dinomaly train	75 (15 $\times$ 5)	60–391 (241.93)	post-hoc sensitivity
VisA	Dinomaly train	60 (12 $\times$ 5)	450–905 (721.58)	post-hoc sensitivity
VisA	PatchCore held-out good	60 (12 $\times$ 5)	90–181 (144.33)	independent holdout

Table 7 closes the data-availability question without changing the confirmatory verdict. It shows that "no Dinomaly train dump" is too broad: MVTec AD and VisA dumps exist, but their checkpoint dependence limits them to exploratory practice sensitivity. Only the PatchCore held-out-good construction supplies an independent calibration pool; no MPDD Dinomaly train dump is available.

On VisA (post-freeze, exploratory, and kept out of the MVTec confirmatory Holm family) the same pooled construction rejects the random-deferral null in all four practice×backbone tests across all five seeds (Holm $p = 2 \times 10^{-3}$), with larger pooled excess-AURC $[0.044, 0.106]$ — the headroom the near-saturated MVTec backbones deny (Table 8, lower block).

Table 8: Pooled excess-AURC audit (C2). Family = {B1 fixed, B2 tuned} × {PatchCore, Dinomaly}, Holm $\alpha = 0.05$. MVTec AD is the frozen confirmatory family: all four tests reject the random-deferral null in every one of the five backbone seeds, and the per-seed values are the frozen confirmatory verdict. VisA (post-freeze) is reported exploratory and is kept out of the MVTec Holm family. Excess-AURC is the seed range; permutation floor $p = 1/2001$.

Benchmark	Backbone	Practice	excess-AURC (seed range)	perm. p	Holm p / reject
MVTec AD	PatchCore	B1 fixed	0.023–0.030	5×10^{-4}	2×10^{-3} / ✓
MVTec AD	PatchCore	B2 tuned	0.036–0.046	5×10^{-4}	2×10^{-3} / ✓
MVTec AD	Dinomaly	B1 fixed	0.045–0.050	5×10^{-4}	2×10^{-3} / ✓
MVTec AD	Dinomaly	B2 tuned	0.027–0.035	5×10^{-4}	2×10^{-3} / ✓
VisA	PatchCore	B1 fixed	0.073–0.089	5×10^{-4}	2×10^{-3} / ✓
VisA	PatchCore	B2 tuned	0.086–0.106	5×10^{-4}	2×10^{-3} / ✓
VisA	Dinomaly	B1 fixed	0.081–0.095	5×10^{-4}	2×10^{-3} / ✓
VisA	Dinomaly	B2 tuned	0.044–0.055	5×10^{-4}	2×10^{-3} / ✓

Verdict: constructive arm on both benchmarks. All four MVTec tests reject in all five seeds (frozen confirmatory); all four VisA tests reject in all five seeds (post-freeze exploratory). Every bootstrap CI excludes 0.

5.16. V1 tier-2 verdict

Under the frozen amendments the tier-2 power floor is applied per repeat, not pooled over repeats (amendment A1). On MVTec AD the escaped-defect axis is per-repeat powered in 13/15 categories (underpowered in toothbrush and transistor, per-repeat $n_{\text{eval,def}} = 15, 20 < 22$). Graded on a per-repeat one-sided 95% Clopper–Pearson upper bound against the 0.13 threshold, the powered cells overwhelmingly fail (115 fail / 15 pass out of 130 powered, 20 excluded as underpowered: toothbrush, transistor; the 15 passes are the near-zero-miss cells — Dinomaly on cable and hazelnut, PatchCore on hazelnut). This is the preregistered expectation of amendment A2, not a certificate failure: split-conformal calibration targets α_{miss} exactly, so a one-sided upper bound at a per-repeat $n \approx 30$–70 sits above the target-plus-3pp tolerance whenever the realized miss rate is near the target. The certificate-validity statistic is tier-1, which passes in all 150 MVTec cells; tier-2 is a stringent estimator-variance readout layered on top (Table A.4 gives the per-category breakdown). The tier-2 false-reject axis is *structurally ungraded* on MVTec AD: it is scored only over G2-certified categories (amendment A3), and those four (cable, hazelnut, screw, transistor) are themselves per-repeat underpowered ($n_{\text{eval,good}} \leq 30 < 36$), so 0 pass / 0 fail out of 150 cells are gradeable (110 G2-refused + 40 underpowered excluded) — exactly the amendment A1+A3 outcome. One caveat we disclose: the power partition and the false-reject buckets are exact (deterministic from the cached counts), but the escaped pass/fail split within powered cells is reconstructed from a per-repeat Clopper–Pearson upper bound built on the pooled rate as the per-repeat point estimate, not from the raw per-repeat records. Across the full 330-cell V1 grid (MVTec AD, VisA, MPDD; two backbones; five seeds), we do not treat the diagnostic pass/fail counts as 330 independent discoveries. They are reported as an engineering uncertainty map with finite-sample exclusions, while inferential discovery control is reserved for the C2 Holm families. On VisA (post-freeze, exploratory) the larger good pools clear the false-reject power floor in part, so that axis is partially gradeable there: escaped 17 pass / 103 fail (of 120), false-reject 4 pass / 66 fail / 50 underpowered (of 120). MPDD is the small-pool endpoint: the same deterministic floors explain the primary-protocol G2 refusal and the train-holdout rescue arm rather than supporting a new confirmatory claim.

5.17. Corruption-drift stress test (post-freeze, exploratory)

The certificates above are within-distribution: they hold under exchangeability between the calibration and evaluation pools. The failure mode a deployed line actually faces is drift, so we stress that assumption directly on a synthetic corruption grid. This analysis is post-freeze and exploratory — it is not preregistered evidence, and we report it because it *bounds* the robustness reading of the frozen certificates rather than confirming it.

Protocol.. For each of the three benchmarks, both backbones, four corruptions (brightness, contrast, Gaussian noise, defocus blur) and three severity levels, we recalibrate the per-category three-way gate on the *clean* repeat-0 stratified calibration half and then route the corresponding *corrupted* test records through those unchanged thresholds. This is exactly the deployment error a plant makes when it calibrates once and the line then shifts. The grid is $3 \times 2 \times 4 \times 3 = 72$ configurations spanning 792 (configuration, category) cells. Alongside the routed decisions we run a drift screen: a two-sample Kolmogorov–Smirnov test, per category, between the clean calibration good-score distribution and the corrupted good-score distribution, with Benjamini–Hochberg adjustment across the categories within each configuration and $\alpha_{\mathrm{KS}} = 0.05$. The screen only observes scores; it never changes a routing decision. We then evaluate the counterfactual overlay in which a detected shift forces refusal.

The screen fires where physics says it should.. The KS screen detects drift in 255/792 cells (32.2%), and detection rises monotonically with severity: 36/264 (13.6%) at level 1, 72/264 (27.3%) at level 2, and 147/264 (55.7%) at level 3. By corruption it is most sensitive to Gaussian noise (111/198, 56.1%) and defocus (69/198, 34.8%), and much weaker on the mild photometric shifts brightness (37/198, 18.7%) and contrast (38/198, 19.2%). The two backbones differ sharply: Dinomaly trips the screen in 164/396 cells (41.4%) against PatchCore's 91/396 (23.0%).

The load-bearing negative result: a good-only monitor does not protect the escaped-defect axis.. Table 9 gives the grid. The screen is genuinely effective on the false-reject axis, which is the axis it observes: of the false-reject exceedances over $\alpha_{\mathrm{fr}} = 0.05$ it catches 51/53 for Dinomaly and 7/13 for PatchCore, leaving only 2/104 (1.9%) and 6/140 (4.3%) exceedances among accepted cells. The escaped-defect axis is the opposite story. Before refusal, realized escaped-defect rates exceed $\alpha_{\mathrm{miss}} = 0.10$ in 135/384 PatchCore G1-eligible cells and 45/384 Dinomaly cells, and the good-only screen catches just 35 and 11 of those respectively; among the cells it accepts, 100/294 (34.0%) PatchCore and 34/223 (15.2%) Dinomaly cells still exceed target. The sharpest cases are accepted PatchCore cells at level-3 Gaussian (11/19 exceed; mean escaped 13.3%, maximum 37.0% on VisA `macaroni1`) and level-3 defocus (13/21; mean 13.1%, maximum 33.3% on MPDD `bracket_white`). The worst accepted false-reject point estimate is 11.9% (PatchCore, VisA `pcb3`, contrast level 3).

Table 9: Corruption-drift stress test (post-freeze exploratory). Thresholds are calibrated on clean data and applied unchanged to corrupted test records. "KS refused" counts the (backbone, category) cells out of 66 whose good-score distribution trips the BH-adjusted KS screen; the two right-hand columns count realized target exceedances ($> \alpha_{\mathrm{miss}} = 0.10$, $> \alpha_{\mathrm{fr}} = 0.05$) among the cells the screen *accepts* and that are certifiable on that axis. Realized point estimates above target are stress evidence about this finite evaluation split, not a disproof of a finite-sample probabilistic guarantee.

Corruption	Severity	KS refused / 66	escaped > 0.10	FR > 0.05
Brightness	1	4	13/60	1/30
Brightness	2	5	12/59	1/27
Brightness	3	28	9/36	1/14
Contrast	1	2	11/62	0/30
Contrast	2	8	16/56	0/25
Contrast	3	28	13/36	2/19
Gaussian	1	23	10/41	0/18
Gaussian	2	41	8/24	0/14
Gaussian	3	47	11/19	1/11
Defocus	1	7	10/57	0/26
Defocus	2	18	8/46	0/20
Defocus	3	44	13/21	2/10

What this does and does not license.. Two caveats bound the reading. First, the KS screen here is heuristic rather than a calibrated type-I-error test: the corrupted good set contains corrupted counterparts of images in the clean calibration sample, so the two samples are dependent while the standard two-sample KS p-value assumes independence. Second, an individual realized rate above target is not by itself a violated guarantee — on the clean held-out split alone, point estimates exceed 0.10 in 12/32 PatchCore and 7/32 Dinomaly eligible strata while the mean clean escaped rates stay below target (8.8% and 7.0%). The level-3 accepted means above 13% with maxima of 33–37% are nonetheless materially worse than that clean baseline. The honest conclusion is therefore twofold: the refusal mechanism generalizes usefully to a detected-drift trigger on the false-reject axis, and a good-score-only monitor is *insufficient* on the escaped-defect axis — with the defect marginal tested and shown not to close the gap either, so the remaining blind spot is joint-distributional (Section 7).

6. Discussion

The engineering result is not that a particular anomaly backbone is uniformly strong; it is that a score-only inspection system can expose the conditions under which automation is warranted, too ambiguous, or statistically uncertifiable. That distinction matters for industrial review because ordinary threshold tuning hides three different failure modes inside a single operating point: escaped defects, rejected good parts, and missing calibration evidence. The dual gate separates those modes. On VisA the larger good pools and lower-ceiling scores allow both axes to certify in all categories; on MVTec AD the escaped-defect side certifies broadly while the false-reject side is mostly pool-size limited; on MPDD the primary protocol refuses G2 outright until an independent train-holdout pool is supplied. A user of the gate therefore receives a deployment requirement — collect more eligible good calibration examples, or accept human review — rather than an uncertified automatic decision.

This refusal discipline is the main difference from the nearest single-threshold baseline. CRC and G1 share the same split-conformal escaped-defect threshold, so Table 3 is not a claim that the proposed gate improves the mathematical G1 guarantee. It shows what is lost when the ambiguous region is collapsed into automatic rejection. On MPDD the same escaped-defect rate is obtained either by rejecting 36.9% of good parts under CRC or by deferring 73.1% of images under the dual gate; VisA and MVTec AD show the same direction with smaller magnitudes. The operational question is then explicit and priced: is human review capacity cheaper than false rejection at the line's economics? In all nine illustrative benchmark–scenario cells the answer is yes under the assumed costs, but the MPDD high-review case is deliberately not oversold — it is a diagnostic operating point unless a plant can staff the review queue.

The uncertainty accounting follows the same conservative principle. The paper reports the full 330-cell V1 grid as a finite-sample uncertainty map, with power exclusions and Clopper–Pearson interval logic rather than treating every pass/fail cell as an independent discovery. The only formal practice-skill discovery claim is the C2 excess-AURC audit, where the family is named in advance, category-stratified permutation tests are paired with category-blocked bootstrap intervals, and Holm adjustment is applied within each seed. That separation is important: V1 certifies the operating policy, C2 asks whether conventional threshold practices contain real deferral skill, and the VisA/MPDD additions test portability of the arithmetic without expanding the frozen MVTec confirmatory family after the fact.

For Results in Engineering readers, the contribution is thus a pre-deployment engineering gate rather than a new anomaly detector. It specifies what data a line must collect, what finite-sample certificate the data support, when the system must refuse to automate, and how to price the human review channel relative to a no-deferral CRC alternative. This is the point at which benchmark experiments become useful to deployment planning: the reported numbers are not claimed to replace a factory validation, but they identify the calibration bottlenecks and cost tradeoffs that a factory validation would need to resolve before certifying a tighter operating point such as $\alpha_{\mathrm{miss}} = 0.01$.

7. Limitations and gated arms

The preregistration was frozen before confirmatory analysis. The confirmatory analysis is MVTec AD; VisA and MPDD are exploratory additions run under the same protocol. The train-holdout remedy is a secondary, explicitly flagged arm, and no uncomputed result is used in the claims. The remaining limitations are scientific rather than bookkeeping: the benchmarks are not production-line data, and the guarantees require calibration pools large enough to clear their finite-sample floors.

The confirmatory audit remains constructive across all five MVTec AD seeds, while the tier-2 readout is reported as an estimator-variance diagnostic rather than a certificate failure. PatchCore's independent train-good holdout supports the secondary G2 remedy; Dinomaly and MPDD do not have an eligible independent holdout for the same promotion claim. The novelty check found no prior method matching the coupled two-axis triage target. Latency measurements show that the added routing and calibration cost is negligible relative to backbone inference (Section 5.9).

Both reproduced backbones sit near ceiling on MVTec AD, leaving the exploratory excess-AURC audit little headroom to detect practice skill there (Section 5.14). VisA was the pre-freeze skeleton's designed answer to exactly this ceiling effect (a lower-ceiling second benchmark, stated there as a plan; VisA itself is not part of the frozen preregistration), and it behaved as designed: the same audit is fully informative on VisA (Section 5.14). The confirmatory pooled audit (Section 5.15) and the post-hoc binding demonstration (Section 5.12) settle the deferral-skill question on MVTec AD directly.

All three benchmarks are academic: no production-line, real-factory, or genuinely time-ordered data backs the industrial framing here (threshold-transfer across production periods is future work), so the paper's claims are about certified triage on the standard inspection benchmarks, not about a deployed line. The exchangeability assumption is stressed only synthetically, by the corruption grid of Section 5.17; a synthetic corruption is not a substitute for real temporal drift.

The corruption-drift stress test (Section 5.17) is the sharpest limitation this paper identifies in its own mechanism. The KS drift screen observes good scores only, so it aligns with false-reject drift and intercepts it well (51/53 Dinomaly and 7/13 PatchCore false-reject exceedances caught). It does not protect the escaped-defect axis: among the cells it accepts, 34.0% of PatchCore and 15.2% of Dinomaly G1-eligible cells still show realized escaped-defect rates above $\alpha_{\text{miss}} = 0.10$, concentrated in severe Gaussian noise and defocus. The obvious remedy — run the same marginal test on defect scores — does not close the gap. We implemented it on the identical frozen inputs, the identical stratified split, and the identical KS + Benjamini–Hochberg machinery (with a reproduction gate: the good-side cells reproduce the frozen screen exactly before the new arm is believed), and the defect-side test intercepts only 5/100 (5.0%) of the accepted-but-exceeding PatchCore cells and 1/34 (2.9%) of the Dinomaly ones (post-freeze exploratory, seed 0). The undetected degradation is therefore not an unmonitored marginal: severe Gaussian and defocus corruptions reshape the *joint* good–defect score relationship — the overlap region and ranking — while moving neither marginal far enough for a marginal two-sample test to fire. We do not claim corruption robustness. The constructive reading is that refusal-on-detected-drift is a sound mechanism whose current instantiation monitors the wrong marginal. The joint-distribution extension is therefore the named remedy, and we implemented and tested it (post-freeze exploratory, seed 0; pairing-respecting Wilcoxon signed-rank per side with a joint max-statistic, Benjamini–Hochberg across the cell family, corrupted images paired 1:1 to their clean counterparts — 104,280/104,280 matched, zero label conflicts; reproduction gates: the frozen defect-marginal catches 5/100 and 1/34 reproduce exactly, and the defect-side KS reproduces on all 792 cells to 10^{-12}). It does not close the blind spot either: the joint monitor catches 88/100 (PatchCore) and 29/34 (Dinomaly) of the accepted-but-exceeding cells, but fires on 88.8% and 92.0% of certified non-exceeding cells respectively — catch rate $\approx$ false-alarm rate, an indiscriminate "any corruption present" detector. Score locations always shift under corruption; what a deployable monitor must detect is the re-ranking of good against defect scores. The tested design space therefore has three levels: marginal tests ($\leq 5\%$ catch), pairing-respecting joint location tests (indiscriminate), and the remaining candidate — a rank-ordering-level statistic (e.g. a permutation test on the gate's certified coverage itself, or an AUC-level statistic across both pools) — which is the required direction

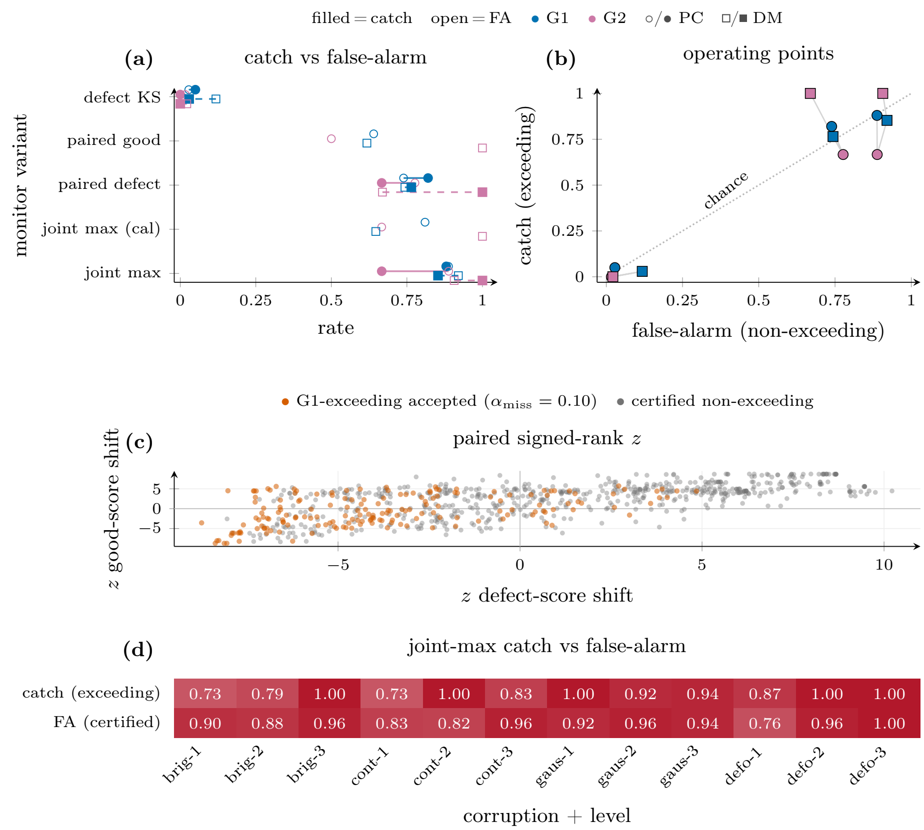

Figure 15: The drift-monitor design space, tested (post-freeze exploratory, seed 0; frozen joint record). **a:** catch rate (filled) vs. false-alarm rate (open) per monitor variant and detector×guarantee arm — the frozen defect-marginal KS catches 5.0%/2.9%; the pairing-respecting variants catch more but at matching false-alarm. **b:** the same arms as operating points against the chance diagonal: no variant separates. **c:** per-cell paired signed-rank z of good- and defect-score shifts; G1-exceeding accepted cells (vermillion) sit inside the same cloud as certified cells — there is no location-only signature to detect. **d:** catch vs. false-alarm rate of the joint max-statistic by corruption and severity level: the two rows track each other at every cell, the indiscriminate regime in one view. From the frozen joint-monitor record (792 cells; reproduction gates passed: defect-marginal catches 5/100 and 1/34 reproduced exactly).

before any drift-facing deployment claim (Figure 15).

Calibrated gates also do not transfer across detectors. The gate is calibrated per detector, and a threshold-transfer experiment (post-freeze exploratory, seed-level) shows that requirement is not cosmetic. Calibrating on PatchCore's calibration-half scores and applying the thresholds to Dinomaly's evaluation-half scores — the identical images, labels, and stratified partition, so the only change is the scoring function — violates the escaped-defect guarantee in 160/165 cells across the three benchmarks (up to +87 points above the α_{miss} tolerance); the reverse direction violates the false-reject guarantee in 80/165 cells. The same-detector diagonal reproduces the frozen certifi-

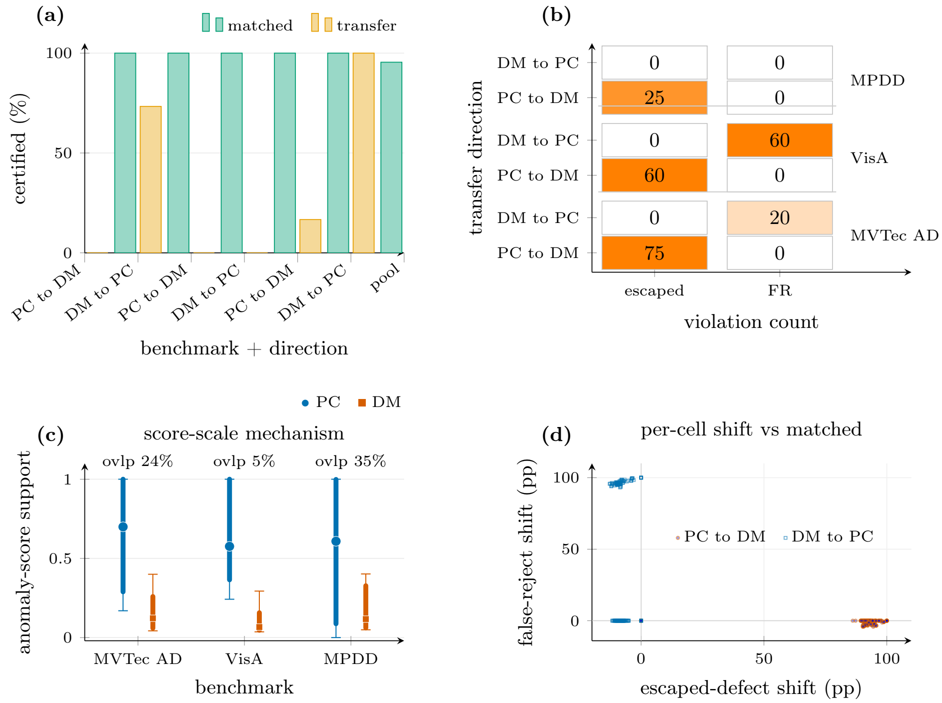

Figure 16: Calibrated gates do not transfer across detectors (post-freeze exploratory, seed-level; frozen transfer record). **a:** certification rate, transfer vs. partition-matched same-detector diagonal, per benchmark and direction — the diagonal certifies nearly everywhere; the transfer arm collapses. **b:** guarantee-violation counts per benchmark and direction: 160/165 escaped-defect violations (PatchCore→Dinomaly, abbreviated "PC to DM" in the figure), 80/165 false-reject violations (reverse, "DM to PC"). **c:** the mechanism — per-detector score-scale ranges barely overlap (support overlap 0.241/0.053/0.353 on MVTec AD/VisA/MPDD), so a threshold calibrated on one detector sits far outside the other's score support. **d:** per-cell realized-rate shift vs. the matched diagonal; the two clusters show total inversion in opposite directions. From the frozen transfer record (330 cells; same-detector diagonal reproduces the frozen certification records exactly, 30/30).

cation records exactly (30/30 cells, with a perturbation negative control), so the failure is the transfer, not the harness. This is a model-shift axis, complementary to — and easier than — the temporal threshold-transfer across production periods named above as future work: even swapping the scoring function on fixed data breaks the certificate. A deployed gate must therefore be recalibrated against the detector it guards, and any detector upgrade re-opens calibration; on VisA the comparison is additionally not architecture-controlled (Dinomaly is a single multi-class model across all twelve categories, PatchCore per-category), so the transfer figures there are descriptive (Figure 16).

Small good-calibration pools are intended behavior, not a shortfall: G2's certifiable count is a direct property of each benchmark's good-pool size, not a conservatism baked into the gate. MVTec AD (4/15), VisA (12/12, identical code path), and MPDD (0/6 under the primary protocol, rescued to

4/6 under the flag-gated train-holdout arm) form three points consistent with pool-size-driven refusal (Sections 5.3, 5.11). MPDD's rescue is PatchCore-only because no MPDD Dinomaly train-score dump exists; the later MVTec/VisA Dinomaly dumps are checkpoint-dependent exploratory sensitivities and do not alter that evidence boundary.

The near-universal G1 certification reported above (15/15 MVTec AD, 12/12 VisA, 5/6 MPDD) holds at $\alpha_{\text{miss}} = 0.10$ — a 10% chance of auto-passing a defective part, well outside what a safety-critical line would accept — and is an artifact of that tolerance being easy to clear against the $\alpha_{\min} = 1/(n_{\text{cal}}^{\text{def}}+1)$ floor at these benchmarks' defect-calibration pools ($n_{\text{cal}}^{\text{def}} \in [15, 70]$ on MVTec AD, Table A.1; $= 50$ on VisA; $\in [7, 36]$ on MPDD). At a deployable tolerance of $\alpha_{\text{miss}} = 0.01$ the floor requires $n_{\text{cal}}^{\text{def}} \geq 99$, which no category on any of the three benchmarks reaches, so G1 refuses in all 33/33 (benchmark, category) cells; at $\alpha_{\text{miss}} = 0.001$ ($n_{\text{cal}}^{\text{def}} \geq 999$) the same is true. The full operating-point frontier is now computed, not just this boundary arithmetic (Figure 5; post-freeze exploratory sweep of $\alpha_{\text{miss}} \in \{0.20, \ldots, 0.01\} \times \alpha_{\text{fr}} \in \{0.10, 0.05, 0.02\}$ on the frozen code path, verified to reproduce every frozen artifact exactly at the paper's operating point): VisA certifies both axes in 12/12 categories down to $\alpha_{\text{miss}} = 0.02$ at modest deferral cost (Dinomaly 6.5% → 12.9%) before the floors bind and certification collapses to 0/12 at 0.01; MVTec AD's both-axis fraction is pinned at 4/15 by the good-pool floors across $\alpha_{\text{miss}} \geq 0.05$ regardless of tolerance; MPDD certifies nothing at any grid point on the primary protocol. The sweep confirms the refusal collapse is exactly where the floor arithmetic predicts, and prices every reachable tighter point. On academic benchmarks of this size, the meaningful demonstration is the refusal discipline itself — that the gate honestly refuses rather than assert an escape guarantee the data cannot support — not a certified industrially-tight escape rate. Certifying escaped-defect at the 1%–0.1% tolerances a safety-critical line would require needs the larger labeled-defective pools (hundreds to thousands per category) that accrue from rework and scrap logging over months of production (Section 3.1), not the few dozen defective images an academic test split provides.

8. Conclusion

We described a pre-deployment three-way triage gate that turns anomaly-detector scores into an auditable benchmark operating decision with finite-sample escaped-defect and false-reject guarantees, and a refusal mechanism that makes "not enough data to certify" a visible outcome. The framework is not yet a factory-validated deployment: the reported operating point is $\alpha_{\text{miss}} = 0.10$, all 33 benchmark categories refuse at 0.01, and production-line drift remains untested. Its engineering value is the explicit calibration-data requirement, priced deferral, and refusal discipline that must be resolved before a line can claim a tighter certificate. On MVTec AD, VisA, and

MPDD, across two backbones and five seeds each, the calibration machinery matches its preregistered arithmetic exactly, with no kill-gate tripped in any of the 30 (backbone, seed) aggregates. The three benchmarks exercise complementary failure modes and together give three points consistent with pool-size-driven refusal ($0/6 \rightarrow 4/15 \rightarrow 12/12$): MVTec AD stresses the refusal side (G2 certifiable in only 4/15 categories under the primary protocol, lifted to 12–13/15 by the frozen train-holdout remedy — PatchCore-only because the eligible independent holdout is unavailable for Dinomaly — with KS-failing categories honestly refused); VisA — lower-ceiling, larger good pools — certifies both axes in all 12 categories, cuts median deferral to 4–19%, and gives the excess-AURC audit the headroom MVTec's saturated backbones deny it; and MPDD (real painted-metal parts, post-freeze exploratory) is the stingy extreme, refusing G2 in all 6 categories under the primary protocol and rescued to 4/6 by the flag-gated train-holdout arm (one category KS-audited out). The confirmatory pooled audit (C2) returns a constructive verdict on MVTec AD — field-standard threshold practice earns real deferral skill over honest random deferral in all five backbone seeds (Section 5.15).

Data availability

MVTec AD [1], VisA [42], and MPDD [29] are publicly available benchmarks. Code and supporting materials for this study are archived at Zenodo (concept DOI: `10.5281/zenodo.21392290`; version 0.4.1) and maintained at `https://github.com/PeterPonyu/inspect-gate`.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI in scientific writing

Generative AI tools were used during manuscript preparation. The author reviewed and edited the resulting content and takes full responsibility for the published article.

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CRediT authorship contribution statement

Zeyu Fu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Appendix A. Supplementary tables and figures

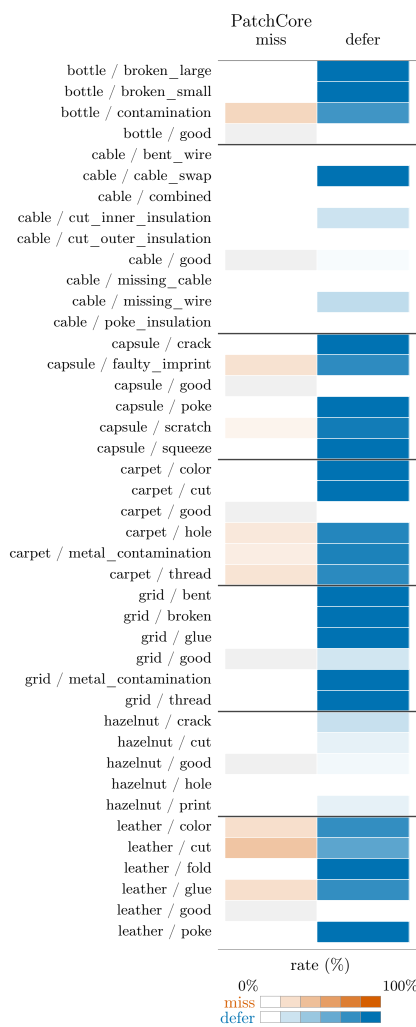

Figure A.1: Defect-type Mondrian map, part 1 (bottle–leather). Per (category, defect-type): realized escaped-defect (miss) rate among defective eval images (left column, vermillion) and deferral rate over all eval images (right column, blue), for both backbones at the operating point (seed 0, repeat 0; post-freeze exploratory). Cell shade encodes the rate; per-cell counts (n_{def}, n) live in the frozen archive. White miss cells mean zero observed misses or an undefined rate when $n_{\mathrm{def}} = 0$. Residual errors concentrate in specific subtle-appearance defect types (e.g. pill::color, leather::cut) rather than uniformly. Horizontal separators mark category boundaries in both backbone panels.

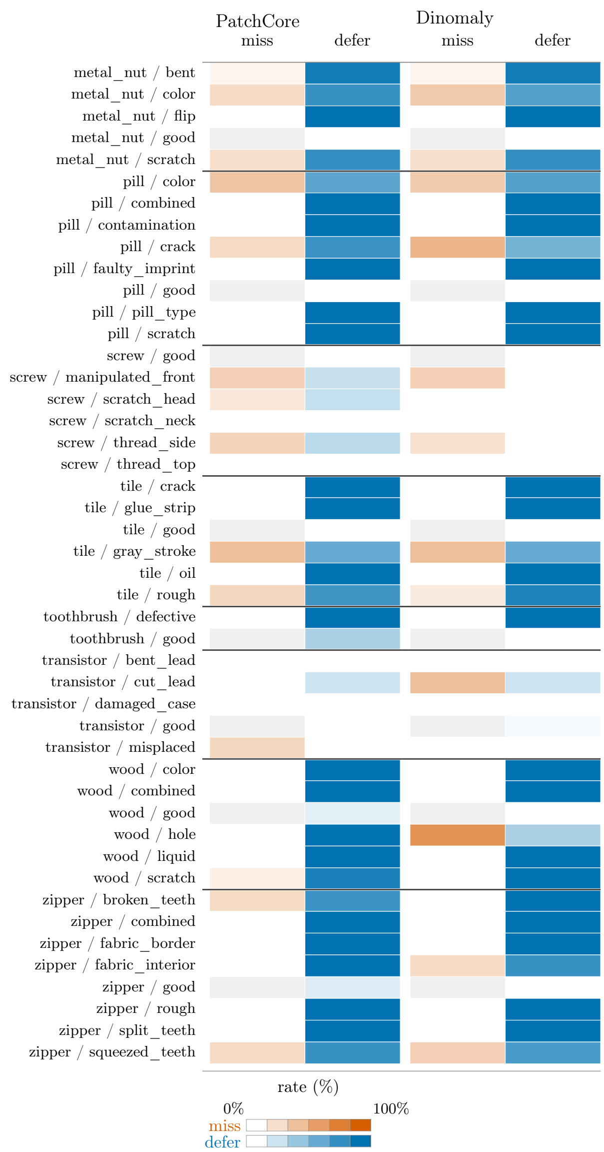

Figure A.2: Defect-type Mondrian map, part 2 (metal_nut–zipper). Encoding as in Figure A.1.

The three appendix tables below were moved out of the main text to keep the Results section readable; every main-text reference to them resolves here. Figures A.1–A.2 give the defect-type (Mondrian) coverage map for MVTec AD (VisA's public ground truth carries a single "bad" stratum per category and MPDD is reported per category).

Table A.1: Per-category certifiability at $\alpha_{\text{miss}} = 0.10$, $\alpha_{\text{fr}} = 0.05$ (backbone-invariant; pure functions of the shared MVTec test split). G1 certifies in 15/15, G2 in 4/15. Source: the frozen preregistration (§3), reproduced with 0/15 mismatch by the analysis pipeline.

Category	$n_{\text{cal}}^{\text{def}}$	$\alpha_{\min}$(G1)	G1@0.10	$n_{\text{cal}}^{\text{good}}$	G2@0.05
bottle	32	0.0303	OK	10	REFUSE
cable	46	0.0213	OK	29	OK
capsule	54	0.0182	OK	12	REFUSE
carpet	44	0.0222	OK	14	REFUSE
grid	28	0.0345	OK	10	REFUSE
hazelnut	35	0.0278	OK	20	OK
leather	46	0.0213	OK	16	REFUSE
metal_nut	46	0.0213	OK	11	REFUSE
pill	70	0.0141	OK	13	REFUSE
screw	60	0.0164	OK	20	OK
tile	42	0.0233	OK	16	REFUSE
toothbrush	15	0.0625	OK	6	REFUSE
transistor	20	0.0476	OK	30	OK
wood	30	0.0323	OK	10	REFUSE
zipper	60	0.0164	OK	16	REFUSE
Certifiable			15/15		4/15

Table A.2: Seed-0 median per-category deferral by benchmark and backbone (Section 5.13), with the mean where computed and the extreme categories. G2 refusal (Section 5.3) empties the auto-reject band and elevates deferral on MVTec AD and MPDD; VisA's universal G2 certification keeps deferral low.

Benchmark	Backbone	Median	Mean	Max category	Min category
MVTec AD	PatchCore	70.0%	55.1%	pill 78.0%	transistor 2.0%
MVTec AD	Dinomaly	69.5%	54.2%	capsule 77.3%	screw 3.1%
VisA	PatchCore	18.7%	—	macaroni2 72.5%	chewinggum 2.0%
VisA	Dinomaly	4.2%	—	capsules 21.9%	candle 2.5%
MPDD	PatchCore	74.1%	—	connector 100%	bracket brown 61.8%
MPDD	Dinomaly	70.9%	—	connector 100%	bracket white 53.3%

Table A.3: Per-backbone companion to Table 3: our dual gate vs. the CRC single-threshold baseline, pooled over five seeds and all categories per backbone (point rates; per-seed min–max bands are included in the published analysis archive). CRC has no deferral mechanism and no G2 certificate by construction.

Benchmark	Backbone	Method	Escaped	False-reject	Deferral
MPDD	PatchCore	Our dual gate	7.1%	0.0%	75.1%
MPDD	PatchCore	CRC	7.1%	41.9%	0.0%
MPDD	Dinomaly	Our dual gate	6.1%	0.0%	71.1%
MPDD	Dinomaly	CRC	6.1%	31.8%	0.0%
VisA	PatchCore	Our dual gate	8.8%	3.7%	26.3%
VisA	PatchCore	CRC	9.7%	28.8%	0.0%
VisA	Dinomaly	Our dual gate	6.5%	2.2%	6.5%
VisA	Dinomaly	CRC	9.3%	3.7%	0.0%
MVTec AD	PatchCore	Our dual gate	7.7%	0.7%	54.8%
MVTec AD	PatchCore	CRC	8.4%	5.2%	0.0%
MVTec AD	Dinomaly	Our dual gate	6.9%	0.3%	53.9%
MVTec AD	Dinomaly	CRC	8.7%	0.9%	0.0%

Table A.4: V1 tier-2 grading, per category (expanded from the aggregate axis totals in the text; every cell out of $10 = 2$ backbones $\times$ 5 seeds). Escaped-defect is pass/fail/unpowered against the per-repeat 0.13 Clopper–Pearson bound (amendment A2); false-reject is pass/fail/unpowered/refused, where "refused" means the category is not G2-certified (amendment A3) and "unpowered" means it is G2-certified but fails the per-repeat power floor (amendment A1). MVTec AD is 0-gradeable on false-reject by construction (no G2-certified category clears the power floor); VisA (post-freeze, exploratory) is partially gradeable.

Category	Escaped-defect (pass/fail/unpwr.)	False-reject (pass/fail/unpwr./refused)
MVTec AD (confirmatory tier-1 family; amendments A1/A2/A3)		
bottle	0/10/0	0/0/0/10
cable	5/5/0	0/0/10/0
capsule	0/10/0	0/0/0/10
carpet	0/10/0	0/0/0/10
grid	0/10/0	0/0/0/10
hazelnut	10/0/0	0/0/10/0
leather	0/10/0	0/0/0/10
metal_nut	0/10/0	0/0/0/10
pill	0/10/0	0/0/0/10
screw	0/10/0	0/0/10/0
tile	0/10/0	0/0/0/10
toothbrush	0/0/10	0/0/0/10
transistor	0/0/10	0/0/10/0
wood	0/10/0	0/0/0/10
zipper	0/10/0	0/0/0/10
VisA (post-freeze, exploratory)		
candle	0/10/0	0/10/0/0
capsules	0/10/0	0/0/10/0
cashew	0/10/0	0/0/10/0
chewinggum	10/0/0	0/0/10/0
fryum	0/10/0	0/0/10/0
macaroni1	0/10/0	0/10/0/0
macaroni2	0/10/0	0/10/0/0
pcb1	0/10/0	0/10/0/0
pcb2	0/10/0	0/10/0/0
pcb3	0/10/0	0/10/0/0
pcb4	5/5/0	4/6/0/0
pipe_fryum	2/8/0	0/0/10/0
Totals		
MVTec AD (150 cells)	15/115/20	0/0/40/110
VisA (120 cells)	17/103/0	4/66/50/0

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